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Master thesis

**On the number of solutions of the
Diophantine equation $|ax^n - by^n| = 1$**

by

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Abstract

The equation $|ax^n - by^n| = 1$, with positive integers a, b and $n \geq 3$, has at most one solution in positive integers x and y . The proof of this result for general $n \geq 5$ relies on techniques from Diophantine approximation (such as rational approximation), the theory of linear forms in logarithms and computational methods. For the case $n = 3$ there is an algebraic method given by Nagell, based on inspecting the coefficients of expansions of numbers of certain forms and using tools from algebraic number theory in cubic fields. With a similar approach, Ljunggren proved the result for $n = 4$. We aim to present the proof of the aforementioned result for $n \neq 4$ following the work of Bennett, de Weger, Nagell and others.

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1 Introduction

A *Thue equation* is an equation of the form

$$F(x, y) = c, \tag{1.1}$$

where $F \in \mathbb{Z}[X, Y]$ is a homogeneous polynomial in two variables with integer coefficients and c is a nonzero integer. These equations are named after the Norwegian mathematician Axel Thue, who showed that if F is irreducible over $\mathbb{Z}[x, y]$ and has degree higher than 2, then equation (1.1) has at most finitely many solutions in integers x and y .

Of particular interest are equations of certain forms, such as equations of the form

$$|ax^n - by^n| = 1, \tag{1.2}$$

with positive coefficients a and b , which are the subject of this thesis. They were studied for $n = 3$ and $b = 1$ by Delone [10], who showed that they have at most one solution in positive integers. A few years later, Nagell [22] independently proved the same result and generalized it by dropping the condition $b = 1$. Using a similar approach, Ljunggren [17] proved the same conclusion in the case $n = 4$.

For arbitrary $n \geq 5$, Domar [11] managed to prove with a few exceptions that equation (1.2) has at most two solutions under certain conditions. His approach was based on the work of Thue [29] and Siegel [27], using sharpened Padé approximation techniques.

Using various Diophantine approximation tools, Bennett and de Weger [3] improved Domar's result, proving that equation (1.2) has at most one solution in positive integers, with some possible exceptions. More precisely, they proved the following theorem.

Theorem 1.1. *If a and b are nonzero integers and $n \geq 3$ is an integer, then the equation (1.2) has at most one solution in positive integers x and y , except possibly when the following conditions hold simultaneously:*

$$\begin{aligned} b &= a + 1, \\ 2 &\leq a \leq \min\{0.3n, 83\}, \\ 17 &\leq n \leq 347. \end{aligned}$$

Later, Bennett [2] handled the exceptions in Theorem 1.1, showing that the theorem holds true in those cases as well. In other words, he proved the following theorem.

Theorem 1.2. *If a and b are nonzero integers and $n \geq 3$ is an integer, then the equation*

$$|ax^n - by^n| = 1 \tag{1.3}$$

has at most one solution in positive integers x and y .

Throughout this thesis we will present the results that led to the proof of this theorem.

In Chapter 2 we introduce some of the main results that are needed for proving Theorem 1.1 and also prove the theorem in some particular cases. We will employ results about Liouville-type estimates and continuous fractions and we will combine them with computational methods.

In Chapter 3 we present a classical result on linear forms in logarithms established by Laurent, Mignotte and Nesterenko in [16]. Using this result allows us to reduce the problem to finitely many cases, which can be treated using the same computational methods seen in Chapter 2.

Chapter 4 provides a short analysis of the method used by Bennett in [2] for inspecting the exceptions of Theorem 1.1. We also present some improvements that allow the programs involved in this proof to run over considerably smaller and fewer numbers, thus reducing the necessary running time.

In Chapters 5 and 6 we delve deeper into the theory underlying some of the main results used in the previous chapters. We present some Liouville-type estimates for irrational numbers, as well as some tools needed for specific constructions that allow us to apply the respective estimates (complex integrals, Padé approximation). As applications of the theory discussed, we present in Chapter 5 the proofs of two important theorems used in Chapter 2, and a sketch of the proof of theorem used in Chapter 4.

Chapter 7 is dedicated to Nagell's proof that the equation

$$Ax^3 + By^3 = 1,$$

with integers A and B has at most one solution in nonzero integers. We only discuss the case $|A|, |B| \neq 1$, because the other case is relatively similar. After providing a brief overview of algebraic number theory focused on the case of real cubic fields, we outline the main results of [22] and give their proofs.

Finally, Chapter 8 provides a brief overview of generalizations to some of the results used in this thesis.

2 The equation $|ax^n - by^n| = 1$ with $n \geq 5$

We begin the study of the equation (1.2) in the case $n \geq 5$, which will be the object of Chapters 2–4. In this chapter, specifically, we introduce a result about a related Diophantine inequality, some Liouville-type estimates and a few classical results on continued fractions. Using these results we will be able to cover a wide range of cases of the equation (1.2) following the work of Bennett and de Weger [3]. The remaining cases will be considered in Chapter 3 using the theory of linear forms in logarithms, as well as in Chapter 4, using some computational techniques involving continuous fractions. We will discuss the cases $n = 3$ and $n = 4$, with a focus on the former, in Chapter 7.

2.1 Preliminary results

We have the following result obtained by Evertse in [13, Theorem 2.1] using approximation methods involving some special polynomials.

Theorem 2.1. *If a, b, c, n are positive integers and $n \geq 3$, then the Diophantine inequality*

$$|ax^n - by^n| \leq c$$

has at most one solution in positive integers satisfying $\gcd(x, y) = 1$ and $\max\{ax^n, by^n\} \geq \mu_n c^{\alpha_n}$, where

$$\begin{aligned} T_n &= 3^{-\frac{n-2}{n}} n \prod_{p|n} p^{\frac{1}{p-1}}, \\ \mu_3 &= T_3^{11/2}, \\ \mu_n &= T_n^{\max\{\frac{n+2}{2(n-3)}, \frac{n}{n-2}\}}, \\ \alpha_3 &= 9, \\ \alpha_n &= \max\left\{\frac{3n-2}{2(n-3)}, \frac{2(n-1)}{n-2}\right\}. \end{aligned}$$

One can show that for $n \geq 5$ we have $\mu_n < n^2$, and therefore $\mu_n < 2^n$. Indeed,

$$\begin{aligned}\mu_n = T_n^{\frac{n}{n-2}} &= \frac{1}{3} \left(n \prod_{p|n} p^{\frac{1}{p-1}} \right)^{\frac{n}{n-2}} \leq \frac{1}{3} \left(2n \prod_{\substack{p|n \\ p \neq 2}} p^{\frac{1}{2}} \right)^{\frac{n}{n-2}} \\ &\leq \frac{1}{3} (2n\sqrt{n})^{\frac{n}{n-2}} = \frac{2^{\frac{n}{n-2}}}{3} \cdot n^{\frac{3n}{2(n-2)}} < n^2.\end{aligned}$$

Thus, by choosing $c = 1$ in Theorem 2.1 we find that our Diophantine equation has at most one solution satisfying $\max\{ax^n, by^n\} \geq 2^n$. Since the last inequality is true for any pair of positive integers $(x, y) \neq (1, 1)$, we shall restrict our attention to the case when $(1, 1)$ is a solution to (1.3), that is when $b = a \pm 1$. It would be enough to consider the case $b = a + 1$, since otherwise we can just swap the roles of x and y . In this case equation (1.3) becomes

$$|ax^n - (a+1)y^n| = 1. \quad (2.1)$$

This equation was solved in [9] in the case $a = 1$ using the modular approach. It was shown that it has the unique solution $x = y = 1$, which we will take for granted, without going into the details of the proof. We will therefore assume $a \geq 2$ from now on.

If (x, y) is a solution to (2.1), then

$$\left| \left(\frac{x}{y} \right)^n - \frac{a+1}{a} \right| = \frac{1}{ay^n}.$$

By the mean value theorem, there is some ξ between x/y and $(a+1)/a$ such that

$$\left| \left(\frac{x}{y} \right)^n - \frac{a+1}{a} \right| = n \left| \frac{x}{y} - \sqrt[n]{\frac{a+1}{a}} \right| \xi^{n-1}.$$

Since $\xi > 1$, it follows that

$$\left| \frac{x}{y} - \sqrt[n]{\frac{a+1}{a}} \right| < \frac{1}{nay^n}. \quad (2.2)$$

Let us define for each integer $n \geq 3$ the number

$$u_n = \prod_{p|n} p^{\frac{1}{p-1}}.$$

We will need the following theorems, whose proofs will be presented in Chapter 5.

Theorem 2.2. *If n and c_n are numbers given in the table below and a is a positive integer such that*

$$\left(\sqrt{a} + \sqrt{a+1} \right)^{2(n-2)} > \left(\frac{nu_n}{c_n} \right)^n,$$

then

$$\left| \sqrt[n]{1 + \frac{1}{a} - \frac{p}{q}} \right| > \frac{1}{a(10^{10}q)^{\lambda_1}},$$

for any positive integers p and q , where

$$\lambda_1 = 1 + \frac{\log \left(\frac{nu_n}{c_n} (\sqrt{a} + \sqrt{a+1})^2 \right)}{\log \left(\frac{c_n}{nu_n} (\sqrt{a} + \sqrt{a+1})^2 \right)}.$$

Table 1

n	c_n	n	c_n	n	c_n	n	c_n
3	2.03	13	1.65	31	1.51	53	1.40
4	1.62	17	1.58	37	1.46	59	1.40
5	1.84	19	1.56	41	1.45	61	1.39
7	1.76	23	1.53	43	1.43	67	1.38
11	1.67	29	1.51	47	1.44	71	1.36

Theorem 2.3. If a and $n \geq 3$ are positive integers such that

$$\left(\sqrt{a} + \sqrt{a+1} \right)^{2(n-2)} > (nu_n)^n,$$

then

$$\left| \sqrt[n]{1 + \frac{1}{a} - \frac{p}{q}} \right| > \frac{1}{8nu_n a q^{\lambda_2}},$$

for any positive integers p and q , where

$$\lambda_2 = 1 + \frac{\log \left(nu_n (\sqrt{a} + \sqrt{a+1})^2 \right)}{\log \left(\frac{1}{nu_n} (\sqrt{a} + \sqrt{a+1})^2 \right)}.$$

Remark 2.4. The expressions of λ_1 and λ_2 have the form

$$1 + \frac{\log u + \log v}{\log u - \log v},$$

where u and v are placeholders for $(\sqrt{a} + \sqrt{a+1})^2$ and $\frac{nu_n}{c_n}$ (or nu_n) respectively. It is easy to see that this is decreasing in the variable u and increasing in the variable v as long as $u < v$. Consequently, for a fixed value of n , we have that the expressions of λ_1 and λ_2 are decreasing in a as long as $(\sqrt{a} + \sqrt{a+1})^2 < \frac{n\mu_n}{c_n}$ or $(\sqrt{a} + \sqrt{a+1})^2 < n\mu_n$, respectively.

Remark 2.5. The hypothesis of either Theorem 2.2 or Theorem 2.3 implies $a \geq 3$.

To see why the second remark is true, we have computed the values of $\left(\frac{nu_n}{c_n}\right)^{\frac{n}{n-2}}$ for $n \leq 19$ given in Table 1, as well as the values of $(nu_n)^{\frac{n}{n-2}}$ for $3 \leq n \leq 9$ and found that in each case they exceed 10. Therefore, in all these cases we have

$$(\sqrt{a} + \sqrt{a+1})^2 > 10,$$

and so $a \geq 3$.

For $n \geq 20$ in the hypothesis of Theorem 2.2 we have

$$\left(\frac{nu_n}{c_n}\right)^{\frac{n}{n-2}} > \frac{n}{2} \geq 10,$$

where he have used the inequalities $u_n > 1$, $c_n < 2$ and $\frac{n}{n-2} > 1$, so we once again have $a \geq 3$.

Finally, for $n \geq 10$ in the hypothesis of Theorem 2.3 we have

$$(nu_n)^{\frac{n}{n-2}} > n \geq 10,$$

so we reach the same conclusion as before.

By combining these theorems with inequality (2.2), we can find upper bounds for y . If the hypotheses of Theorem 2.2 are met, then we obtain

$$\frac{1}{nay^n} > \frac{1}{a(10^{10}y)^{\lambda_1}},$$

which is equivalent to

$$y^{n-\lambda_1} < \frac{10^{10\lambda_1}}{n}.$$

If $\lambda_1 < n$, then this implies $y < \left(\frac{10^{10\lambda_1}}{n}\right)^{\frac{1}{n-\lambda_1}}$.

On the other hand, one can see that the condition $\lambda_1 < n$ actually implies the condition in Theorem 2.2. Indeed, if we assume that

$$\left(\sqrt{a} + \sqrt{a+1}\right)^{2(n-2)} \leq \left(\frac{nu_n}{c_n}\right)^n,$$

then using Remark 2.4, we get

$$\lambda_1 \leq 1 + \frac{\log\left(\frac{nu_n}{c_n} \cdot \left(\frac{nu_n}{c_n}\right)^{\frac{n}{n-2}}\right)}{\log\left(\frac{c_n}{nu_n} \cdot \left(\frac{nu_n}{c_n}\right)^{\frac{n}{n-2}}\right)} = n.$$

We have therefore just proved the following lemma.

Lemma 2.6. *If $\lambda_1 < n$ (as defined in Theorem 2.2), then $y < \left(\frac{10^{10\lambda_1}}{n}\right)^{\frac{1}{n-\lambda_1}}$.*

By complete analogy, we have the following.

Lemma 2.7. *If $\lambda_2 < n$ (as defined in Theorem 2.3), then $y < (8\mu_n)^{\frac{1}{n-\lambda_2}}$.*

2.2 Particular cases of the equation

Before making any advancement, we establish the following elementary, yet useful, result.

Lemma 2.8. *If (x, y) is a positive solution to (2.1) then $x = y = 1$ or $x > y > an$.*

Proof. Let $(x, y) \neq (1, 1)$ be a positive solution to (1.2) and let's assume that $x \leq y$. Then $y > 1$ and we have

$$1 = |ax^n - (a+1)y^n| \geq (a+1)y^n - ax^n \geq y^n > 1,$$

a contradiction.

Therefore $x > y$, and so we have

$$\begin{aligned} 1 &= |ax^n - (a+1)y^n| \geq ax^n - (a+1)y^n \\ &\geq a(y+1)^n - (a+1)y^n \\ &= -y^n + any^{n-1} + a \sum_{k=2}^n \binom{n}{k} y^k \\ &> -y^n + any^{n-1} + 1. \end{aligned}$$

Consequently $y^n > any^{n-1}$, so $y > an$. □

We will also need the following two classical results about continuous fractions. Their proofs can be found in the mentioned references or in almost any source that covers continuous fractions.

Theorem 2.9 ([15], Proposition 8.7). *Suppose that ξ is irrational and that a/b is rational with*

$$\left| \xi - \frac{a}{b} \right| < \frac{1}{2b^2}.$$

Then a/b is a continued fraction convergent to ξ .

Theorem 2.10 ([12], Proposition 1.11). *Let ξ be a real number. If p_n/q_n is the n -th convergent of the continued fraction expansion of ξ and a_{n+1} is the $(n+1)$ -st partial quotient, then*

$$\left| \xi - \frac{p_n}{q_n} \right| > \frac{1}{(a_{n+1} + 2)q_n^2}.$$

Returning to our equation, we have the following lemma.

Lemma 2.11. *If a and n are positive integers with $n \geq 5$ and $a \geq 0.3n$ then equation (2.1) has the unique positive solution $(x, y) = (1, 1)$.*

Proof. It is clearly enough to prove this lemma for prime values of n only. In this case we have $u_n = n^{\frac{1}{n-1}}$.

From $a \geq 0.3n$ we get

$$(\sqrt{a} + \sqrt{a+1})^2 > 4a \geq 1.2n.$$

If $n \geq 79$, then we easily obtain $nu_n < 1.06n$ and, using Remark 2.5, we get

$$\lambda_2 < 1 + \frac{\log(nu_n \cdot 1.2n)}{\log\left(\frac{1}{nu_n} \cdot 1.2n\right)} = \frac{2\log(1.2n)}{\log\left(\frac{1}{nu_n} \cdot 1.2n\right)} < \frac{2\log(1.2n)}{\log \frac{1.2}{1.06}} < n - 4.$$

Furthermore, by using Lemma 2.7 and the inequality $n^{\frac{1}{n-1}} < 2$, we obtain

$$y < 16^{\frac{1}{n-\lambda_2}} < 16^{\frac{1}{4}} = 2,$$

and so $y = 1$ and therefore $x = 1$.

For each $n \in \{59, 61, 67, 71, 73\}$ we verify with a computer program (see `code6.txt` in [30]) that the hypothesis of Lemma 2.7 is satisfied for $a = \lceil 0.3n \rceil$ and, moreover, that $(8u_n)^{\frac{1}{n-\lambda_2}} < an$. Consequently, in each of these cases we have $y < an$. This is in fact enough to guarantee that $y < an$ holds for any $a \geq 0.3n$ because, as the expression of λ_2 is decreasing in a , then so is the expression of $(8u_n)^{\frac{1}{n-\lambda_2}}$, whereas an is increasing in a . We are therefore done in these cases due to Lemma 2.8.

The same observation allows us to find minimal values a_0 such that for any $a \geq a_0$ the hypothesis of Lemma 2.7 is satisfied and $(8\mu_n)^{\frac{1}{n-\lambda_2}} < an$. These are listed in the following table.

n	5	7	11	13	17	19	23	29	31	37	41	43	47	53
a_0	10	7	7	7	8	8	9	11	11	13	14	14	15	17

We are therefore done in the cases (n, a) with $a \geq a_0$ thanks to Lemma 2.8.

For the remaining values of a we aim to use Lemma 2.6. We find that it is applicable in the cases presented in the table below and produces the upper bounds c_0 for y .

Table 2

n	a	c_0	n	a	c_0
5	$3 \leq a \leq 9$	10^{59}	29	$9 \leq a \leq 10$	10^8
7	$3 \leq a \leq 6$	10^{43}	31	10	10^7
11	$4 \leq a \leq 6$	10^{22}	37	12	10^6
13	$4 \leq a \leq 6$	10^{27}	41	13	10^6
17	$6 \leq a \leq 7$	10^{12}	43	13	10^6
19	$6 \leq a \leq 7$	10^{14}	47	—	—
23	$7 \leq a \leq 8$	10^{12}	53	16	10^5

Notice that the case $n = 47$ was fully inspected before; this is why it is empty in the second table.

Let us notice that $\frac{1}{nay^n} < \frac{1}{2n^2}$, so Theorem 2.9 implies that x/y is a convergent of $\sqrt[n]{1+1/a}$, say $(x, y) = (p_k, q_k)$, where k is a positive integer and p_k/q_k is the k -th convergent of $\sqrt[n]{1+1/a}$. By Theorem 2.10 we have

$$\left| \sqrt[n]{1 + \frac{1}{a}} - \frac{p_k}{q_k} \right| > \frac{1}{(a_{k+1} + 2)q_k^2},$$

where a_{k+1} is the $(k+1)$ -th partial quotient of the continued fraction expansion of $\sqrt[n]{1+1/a}$. Now inequality (2.2) implies

$$a_{k+1} \geq anq_k^{n-2} - 1. \quad (2.3)$$

For each pair (n, a) in Table 2 we test the first five convergents and find that they do not produce any solution, except for $(1, 1)$. We also find in every case that $q_5 \geq 151$, which means that $q_k \geq 151$, for any $k \geq 5$, and so by (2.3) a solution to (2.1), other than $(1, 1)$, would produce a partial quotient exceeding 10^7 . By calculating all the partial quotients for the convergents whose denominator do not exceed the bound indicated in Table 2, we do not find partial quotients larger than 10^7 , which proves that no other solutions exist. $\square$

Before moving on, we will make a few comments regarding these tables. It turns out that there are some minor errors in the data provided by Bennett and de Weger. The indicated values of a_0 corresponding to $n = 5$ and $n = 7$ are 7 and 6, respectively. One can easily verify that those values do not actually work, but ours do. Consequently, the values they consider for a in the second table corresponding to 5 and 7 are $3 \leq a \leq 6$ and $3 \leq a \leq 5$, respectively. This does not affect the bound c_0 because the expression of the upper bound given by Lemma 2.6 is decreasing in a (over the range where the lemma is applicable), and so the value of c_0 is determined by the smallest a in each case.

Another thing to note is that they include the value $a = 3$ in the case $n = 11$ and so their c_0 in this case is 10^{158} . This case is not part of Lemma 2.11 as the condition $a \geq 0.3n$ is not satisfied, but it is nonetheless necessary for proving Theorem 1.1. Lemma 2.6 is still applicable in this case, so the case can be indeed worked out with the method presented above.

The cases corresponding to $a = 2$ and $a = 3$ which are not covered by Lemma 2.11 are treated separately in [3] using techniques from computational algebraic number theory, thus leading to the following lemma.

Lemma 2.12. *If a and n are positive integers with $5 \leq n \leq 16$, then equation (2.1) has the unique positive solution $(x, y) = (1, 1)$.*

Lemma 2.13. *If $a \geq 84$ and $n < 331$, then the equation (2.1) has the unique positive solution $(x, y) = (1, 1)$.*

Proof. As established in Remark 2.4, the expression of λ_2 from Theorem 2.3 is decreasing in a in the range of a for which $(\sqrt{a} + \sqrt{a+1})^2 > n\mu_n$. One easily checks that

$$(\sqrt{a} + \sqrt{a+1})^2 \geq (\sqrt{84} + \sqrt{85})^2 > 337 > n\mu_n,$$

so we can conclude that

$$\lambda_2 \leq 1 + \frac{\log \left(n^{\frac{n}{n-1}} (\sqrt{84} + \sqrt{85})^2 \right)}{\log \left(n^{-\frac{n}{n-1}} (\sqrt{84} + \sqrt{85})^2 \right)}.$$

We are now going to prove that the right hand side in the last relation is smaller than n . Using a computer program (see `code4.txt` in [30]), we find that the minimum of $n - \lambda_2$ for the values under consideration is about 0.789 (achieved for $n = 3$). Hence $n - \lambda_2 > 0.7$, and so we can apply Lemma 2.7, thus getting $y < \left(8n^{\frac{1}{n-1}} \right)^{\frac{1}{n-\lambda_2}}$.

By basic calculus, we have $n^{\frac{1}{n-1}} \leq \sqrt{3}$ for $n \geq 3$, and so

$$\left(8n^{\frac{1}{n-1}} \right)^{\frac{1}{n-\lambda_2}} \leq \left(8\sqrt{3} \right)^{\frac{1}{n-\lambda_2}} < \left(8\sqrt{3} \right)^{\frac{10}{7}} < 43 < an,$$

which completes the proof in view of Lemma 2.8. $\square$

2.3 Preparation for the next chapter

In order to prove Theorem 1.1 when $2 \leq a \leq 83$, we have to consider $n \geq 349$, and when $a \geq 84$, in view of Lemma 2.13 it is enough to consider $n \geq 331$. Notice that in the first case we have $x > y > 2 \cdot 349$, and so $x \geq 700$, whereas in the second case we have $x > 84 \cdot 331$. Either way, $x \geq 700$. We also have

$$x > \frac{a^2}{0.3} > (a+1)^2.$$

By rewriting equation (2.1) as

$$\left| 1 - \left(1 + \frac{1}{a} \right) \left(\frac{y}{x} \right)^n \right| = \frac{1}{ax^n},$$

we obtain

$$\left(1 + \frac{1}{a} \right) \left(\frac{y}{x} \right)^n = 1 \pm \frac{1}{ax^n}, \tag{2.4}$$

and so

$$\log \left(1 + \frac{1}{a} \right) - n \log \frac{x}{y} = \log \left(1 \pm \frac{1}{ax^n} \right).$$

Using the inequality $|\log(1+t)| < 1.01|t|$ for $t \in (-0.01, \infty) \setminus \{0\}$, we deduce

$$\left| \log \left(1 + \frac{1}{a} \right) - n \log \frac{x}{y} \right| < \frac{1.01}{ax^n}. \tag{2.5}$$

It should be noted that in [3] the authors are using inequality (2.5) with the constant 1 instead of 1.01. This, however, fails to be true for the minus sign choice in (2.4), since

for $t \in (-1, 0)$ we have $|\log(1+t)| > |t|$. By changing the right hand side of the last inequality to $1.01|t|$, we are able to flip the sign of the inequality for t negative with small absolute value, and thus get a correct upper bound.

We might also want to obtain a lower bound for the left hand side of (2.5) in order to deduce a new inequality. We will see in the next chapter how to find such a lower bound.

3 Linear forms in logarithms

A linear form in logarithms of algebraic numbers is an expression of the form

$$b_1 \log \alpha_1 + \dots + b_m \log \alpha_m,$$

where $\alpha_1, \dots, \alpha_m$ and $b_1, \dots, b_m$ are algebraic numbers.

A central problem concerning such expressions is to establish explicit lower bounds for their absolute value when this is nonzero. Such a bound was obtained for the first time in 1935 by Gel'fond [14] for $m = 2$. In 1966 Baker [1] provided a bound for general n , and several improvements were published over the time.

Linear forms in logarithms are particularly helpful in solving some Diophantine equations when the coefficients b_i are assumed to be integers. In practice one may also wish to consider results about expressions involving only two or three logarithms. Such an example is a result established by Laurent, Mignotte and Nesterenko in [16], which we will use throughout this chapter in order to complete the proof of Theorem 1.1.

3.1 Main theorem

Before stating the main result, we recall the definition of the absolute logarithmic height of an algebraic number α . If $a(x - \alpha_1) \dots (x - \alpha_d)$ is the minimal polynomial of α over $\mathbb{Z}$ with $a > 0$, then

$$h(\alpha) := \frac{1}{d} \left(\log a + \sum_{i=1}^d \log \max \{1, |\alpha_i|\} \right)$$

is called the *absolute logarithmic height* of α . In particular, if $\alpha = u/v$, with u and v coprime integers, then $h(\alpha) = \log \max\{|u|, |v|\}$.

Theorem 3.1 ([16], Théorème 3). *Let α_1 and α_2 be two positive real algebraic numbers such that $\log \alpha_1$ and $\log \alpha_2$ are linearly independent over $\mathbb{Q}$. Let b_1 and b_2 be positive integers and*

$$\Lambda := b_2 \log \alpha_2 - b_1 \log \alpha_1.$$

Let $D = [\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}]$ and for any $\rho > 1$, consider $\lambda = \log \rho$ and a_1, a_2 and h such

that

$$\begin{aligned} a_i &\geq (\rho - 1)|\log \alpha_i| + 2Dh(\alpha_i), \quad i \in \{0, 1\} \\ a_1 + a_2 &\geq 4 \max\{1, \lambda\}, \\ \frac{1}{a_1} + \frac{1}{a_2} &\leq \min\{1, \lambda^{-1}\}, \\ h &\geq \max \left\{ \frac{D}{2}, 5\lambda, D \left(\log \left(\frac{b_1}{a_2} + \frac{b_2}{a_1} \right) + \log \lambda + 1.56 \right) \right\}, \end{aligned}$$

where $\lambda = \log \rho$. Then

$$\begin{aligned} \log |\Lambda| &\geq -\frac{a_1 a_2}{9\lambda} A^2 - \frac{2}{3}(a_1 + a_2)A - \frac{16}{3}\sqrt{2a_1 a_2} B^{3/2} - \\ &\quad - \log \left(\frac{a_1 a_2}{\lambda} B^2 \right) - \frac{3}{2}\lambda - 2h - \frac{3}{20}, \end{aligned}$$

where

$$A = \frac{4h}{\lambda} + 4 + \frac{\lambda}{h} \text{ and } B = 1 + \frac{h}{\lambda}.$$

Remark. As noted in [20], in the original paper this result is stated with the stronger condition

$$\min\{a_1, a_2\} \geq \max\{2, 2\lambda\}$$

instead of

$$\begin{aligned} a_1 + a_2 &\geq 4 \max\{1, \lambda\}, \\ \frac{1}{a_1} + \frac{1}{a_2} &\leq \min\{1, \lambda^{-1}\}. \end{aligned}$$

However, the proof only uses these last two inequalities.

3.2 Setup for the application

We recall an objective announced in section 2.3: finding a lower bound for

$$\left| \log \left(1 + \frac{1}{a} \right) - n \log \frac{x}{y} \right|.$$

To achieve this, we might want to apply Theorem 3.1 with the obvious choices $\alpha_1 = x/y$, $\alpha_2 = 1 + 1/a$, $b_1 = n$ and $b_2 = 1$. Notice that α_1 and α_2 are rational, so for these choices we have $D = 1$. Moreover, we have the following result, which ensures that the choices of α_1 and α_2 are valid.

Proposition. *The numbers $\log \alpha_1$ and $\log \alpha_2$ are linearly independent over $\mathbb{Q}$.*

Proof. Let $u, v \in \mathbb{Q}$ be such that $u \log \alpha_1 + v \log \alpha_2 = 0$. By scaling, we can in fact assume that u and v are integers and $u \geq 0$. Since $\log \alpha_1$ and $\log \alpha_2$ are positive, it follows that $v \leq 0$. We have $\alpha_1^u = \alpha_2^{-v}$, or equivalently, $x^u a^{-v} = y^u (a+1)^{-v}$. On the other hand, equation (1.3) implies $\gcd(x, y) = 1$ and $\gcd(x, a+1) = 1$, and so we must have $u = 0$, and therefore $v = 0$ as well. $\square$

Following [20], we take

$$\rho = \begin{cases} 5.8, & \text{if } a = 2 \\ 1 + \frac{\log(a+1)}{\log(1+\frac{1}{a})}, & \text{if } a \geq 3 \end{cases}.$$

For this choice of ρ we have

$$\lambda = \log \rho = \log \left(1 + \frac{\log(a+1)}{\log(1+\frac{1}{a})} \right) > \log(1 + a \log(a+1)) > \log(a+1)$$

if $a \geq 3$. By direct computation, we see that the inequality $\lambda > \log(a+1)$ is also true for $a = 2$.

We will now find an upper bound for λ of the form $\alpha \log(a+1)$. To do this, we have to bound the expression

$$E(a) := \frac{\log \left(1 + \frac{\log(a+1)}{\log(1+\frac{1}{a})} \right)}{\log(a+1)}$$

from above.

Since $\log(1 + \frac{1}{t}) > \frac{1}{t+0.5}$, for any $t > 0$, we get

$$\begin{aligned} E(a) &< \frac{\log(1 + (a+0.5) \log(a+1))}{\log(a+1)} \\ &= \frac{\log(a+0.5)}{\log(a+1)} + \frac{\log \log(a+1)}{\log(a+1)} + \frac{\ln \left(1 + \frac{1}{(a+1) \log(a+1)} \right)}{\log(a+1)} \\ &< \frac{\ln(a+0.5)}{\ln(a+1)} + \frac{\log \log(a+1)}{\log(a+1)} + \frac{1}{(a+1) \log^2(a+1)}. \end{aligned}$$

It is easy to check that the expression above is decreasing in a for $a \geq 15$, and so

$$E(a) \leq \frac{\log 15.5}{\log 16} + \frac{\log \log 16}{\log 16} + \frac{1}{16 \log^2 16} < 1.365,$$

if $a \geq 15$. For $a \in \{3, \dots, 14\}$, by computing the values of $E(a)$, we find that the inequality $E(a) < 1.365$ is still true.

We therefore have

$$\log(a+1) < \lambda < 1.365 \log(a+1). \quad (3.1)$$

We want to find a_1 and a_2 satisfying

$$\begin{aligned} a_1 &\geq (\rho - 1) \log \frac{x}{y} + 2 \log x, \\ a_2 &\geq (\rho - 1) \log \left(1 + \frac{1}{a}\right) + 2 \log(a + 1), \end{aligned}$$

or equivalently,

$$\begin{aligned} a_1 &\geq \frac{\log(a + 1)}{\log \left(1 + \frac{1}{a}\right)} \log \frac{x}{y} + 2 \log x, \\ a_2 &\geq 3 \log(a + 1). \end{aligned}$$

We will therefore choose

$$a_2 = 3 \log(a + 1).$$

Now, from inequality (2.5) we deduce

$$\log \frac{x}{y} < \frac{1}{n} \left(\log \left(1 + \frac{1}{a}\right) + \frac{1.01}{ax^n} \right),$$

and so

$$\begin{aligned} \frac{\log(a + 1)}{\log \left(1 + \frac{1}{a}\right)} \log \frac{x}{y} &< \frac{\log(a + 1)}{n} + \frac{1.01 \log(a + 1)}{nax^n \log \left(1 + \frac{1}{a}\right)} \\ &< \frac{\log x}{2n} + \frac{1.01 \log x}{2nx^n} \\ &\leq \left(\frac{1}{662} + \frac{1.01}{662 \cdot 700^{331}} \right) \log x \\ &< 0.002 \log x. \end{aligned}$$

We can therefore choose

$$a_1 = 2.002 \log x.$$

In order to make sure our choices are valid, we still have to check if the rest of the conditions are satisfied, that is, we have to prove the inequalities

$$\begin{aligned} a_1 + a_2 &\geq 4 \max\{1, \lambda\}, \\ \frac{1}{a_1} + \frac{1}{a_2} &\leq \min\{1, \lambda^{-1}\}. \end{aligned}$$

Since $\lambda > 1$, these inequalities become

$$\begin{aligned} a_1 + a_2 &\geq 4\lambda, \\ \frac{1}{a_1} + \frac{1}{a_2} &\leq \lambda^{-1}. \end{aligned}$$

By using the inequalities $x > (a+1)^2$ and (3.1), we have

$$\begin{aligned}
a_1 + a_2 &= 2.002 \log x + 3 \log(a+1) \\
&> 4.004 \log(a+1) + 3 \log(a+1) \\
&= 7.004 \log(a+1) \\
&> \frac{7.004}{1.365} \lambda \\
&> 4\lambda,
\end{aligned}$$

so the first inequality holds. As for the second one, let us notice that from (3.1) it would be enough to prove

$$\frac{1}{a_1} + \frac{1}{a_2} \leq \frac{1}{1.365 \log(a+1)},$$

which is equivalent to

$$\frac{1}{2.002 \log x} \leq \left(\frac{1}{1.365} - \frac{1}{3} \right) \frac{1}{\log(a+1)}.$$

Using the inequality $x > (a+1)^2$, we immediately see that the inequality above is true.

Finally, we have to find a value for h satisfying

$$h \geq \max \left\{ 5\lambda, \log \left(\frac{n}{3 \log(a+1)} + \frac{1}{2.002 \log x} + \log \lambda + 1.56 \right) \right\}.$$

Using (3.1) and the inequalities $x \geq 700$ and $a \geq 0.3n$, we obtain

$$\begin{aligned}
&\log \left(\frac{n}{3 \log(a+1)} + \frac{1}{2.002 \log x} \right) + \log \lambda + 1.56 \\
&< \log \left(\frac{n}{3 \log(a+1)} + \frac{1}{2.002 \log 700} \right) + \log(1.365 \log(a+1)) + 1.56 \\
&= \log n + \log \left(1 + \frac{3 \log(a+1)}{2.002n \log 700} \right) + \log 1.365 - \log 3 + 1.56 \\
&< \log n + \frac{3 \log(a+1)}{2.002n \log 700} + 0.773 \\
&< \log n + \frac{3 \log(0.3n+1)}{2.002n \log 700} + 0.773 \\
&< \log n + 0.01 \log n + 0.14 \log n \\
&= 1.15 \log n.
\end{aligned}$$

Therefore we can choose

$$h = \max\{5\lambda, 1.15 \log n\}.$$

To recapitulate, we have proved that the following are valid choices for the variables in the statement of Theorem 3.1:

$$\begin{aligned}
\alpha_1 &= \frac{x}{y}, \\
\alpha_2 &= 1 + \frac{1}{a}, \\
b_1 &= n, \\
b_2 &= 1, \\
\rho &= \begin{cases} 5.8, & \text{if } a = 2 \\ 1 + \frac{\log(a+1)}{\log(1+\frac{1}{a})}, & \text{if } a \geq 3 \end{cases}, \\
\lambda &= \log \rho, \\
a_1 &= 2.002 \log x, \\
a_2 &= 3 \log(a+1), \\
h &= \max\{5\lambda, 1.15 \log n\}.
\end{aligned}$$

We will now apply Theorem 3.1 for the aforementioned variables. For this we need an explicit value of h , so we will consider each of the cases $5\lambda < 1.15 \log n$ and $5\lambda \geq 1.15 \log n$.

3.3 The case of large n

We assume $5\lambda < 1.15 \log n$, so $h = 1.15 \log n$. In this case we have

$$n > e^{\frac{5}{1.15}\lambda} > (a+1)^{4.3}$$

and at the same time

$$n > e^{\frac{5}{1.15}\lambda} > e^{\frac{5}{1.15} \log 5.8} > 2000.$$

Using these inequalities we obtain the following bounds for A and B (defined in Theorem 3.1):

$$\begin{aligned}
A &= \frac{4.6 \log n}{\log 5.8} + 4 + \frac{\log 5.8}{1.15 \log n} < 3.2 \log n, \\
B &< \frac{A}{4} < 0.8 \log n.
\end{aligned}$$

Using these bounds and Theorem 3.1, for $a = 2$ we obtain

$$\begin{aligned}
\log |\Lambda| &> -\frac{2.002 \log 3 \log x}{3 \log 5.8} \cdot 10.24 (\log n)^2 - \frac{2}{3} (2.002 \log x + 3 \log 3) (3.2 \log n) - \\
&\quad - \frac{12.8}{3} \sqrt{9.6096 \log 3} \cdot (\log n)^{3/2} (\log x)^{1/2} - \\
&\quad - \log \left(\frac{6.006 \log 3 \log x}{\log 5.8} \cdot 0.64 (\log n)^2 \right) - 1.5 \log 5.8 - 2.3 \log n - 0.15,
\end{aligned}$$

which, after adequate rounding, implies

$$\begin{aligned} \log |\Lambda| &> - \left(4.3(\log n)^2 + 4.3 \log n + 9.4 \frac{\log n}{\log x} + \right. \\ &\quad \left. + 13.9 \frac{(\log n)^{3/2}}{(\log x)^{1/2}} + \frac{\log \log x}{\log x} + \frac{2 \log \log n}{\log x} + \frac{3.7}{\log x} \right) \log x. \end{aligned}$$

Recalling that $x \geq 700$, we further get

$$\log |\Lambda| > - \left(4.3(\log n)^2 + 5.8 \log n + 5.5(\log n)^{3/2} + 0.31 \log \log n + 0.9 \right) \log x.$$

On the other hand, inequality (2.5) implies

$$\log |\Lambda| < -n \log x, \quad (3.2)$$

and consequently

$$n < 4.3(\log n)^2 + 5.8 \log n + 5.5(\log n)^{3/2} + 0.31 \log \log n + 0.9. \quad (3.3)$$

Since $n > 2000$, the following inequalities hold:

$$(\log n)^2 < 0.03n, \quad (3.4)$$

$$\log n < 0.01n, \quad (3.5)$$

$$(\log n)^{3/2} < 0.02n, \quad (3.6)$$

$$\log \log n < 0.1n, \quad (3.7)$$

$$0.9 < 0.1n. \quad (3.8)$$

Using these inequalities in (3.3) yields a contradiction.

We now assume $a \geq 3$. Since $h > 5\lambda$, we get

$$\begin{aligned} A &= \frac{4h}{\lambda} + 4 + \frac{\lambda}{h} = \frac{h}{\lambda} \left(4 + 4 \cdot \frac{\lambda}{h} + \left(\frac{\lambda}{h} \right)^2 \right) < 4.84 \frac{h}{\lambda} < 5.6 \frac{\log n}{\lambda}, \\ B &< \frac{A}{4} < 1.4 \frac{\log n}{\lambda}. \end{aligned}$$

Using Theorem 3.1 we get

$$\begin{aligned} \log |\Lambda| &> - \frac{2.002 \log(a+1) \log x}{3} \cdot \left(\frac{5.6 \log n}{\lambda} \right)^2 - \\ &\quad - \frac{2}{3} (2.002 \log x + 3 \log(a+1)) \cdot \frac{5.6 \log n}{\lambda} - \\ &\quad - \frac{16}{3} \sqrt{12.012 \log(a+1) \log x} \cdot (1.4)^{3/2} \cdot (\log n)^{3/2} \cdot \lambda^{-3/2} - \\ &\quad - \log \left(\frac{6.006 \log(a+1) \log x}{\lambda} \cdot \left(\frac{1.4 \log n}{\lambda} \right)^2 \right) - \\ &\quad - \frac{3}{2} \lambda - 2.3 \log n - 0.15. \end{aligned}$$

Since $2 \log(a+1) < \log x$, this implies

$$\begin{aligned} \log |\Lambda| &> -\frac{2.002 \cdot (5.6)^2 \log^2 n \log x}{3} \lambda^{-2} - \frac{4.004}{3} \log x \cdot \frac{5.6 \log n}{\lambda} - \\ &\quad - \frac{16}{3} \sqrt{12.012 \log x \cdot (1.4)^{3/2} \cdot (\log n)^{3/2} \cdot \lambda^{-1}} - \\ &\quad - \log \left(6.006 \log x \cdot \left(\frac{1.4 \log n}{\lambda} \right)^2 \right) - 13.5 \log n - 0.15, \end{aligned}$$

which, after using that $\lambda > 1.76$ and rounding, implies

$$\begin{aligned} \log |\Lambda| &> -6.8 \log^2 n \log x - 4.3 \log n \log x - 13.5 \log n - \\ &\quad - 17.5 (\log n)^{3/2} (\log x)^{1/2} - 2 \log \log n - \log \log x - 1.6. \end{aligned}$$

Since $x \geq 700$, the last inequality implies

$$\log |\Lambda| > (-6.8 \log^2 n - 6.4 \log n - 6.9 (\log n)^{3/2} - 0.4 \log \log n - 0.6) \log x,$$

which along with (3.2) yields

$$n < 6.8 \log^2 n + 6.4 \log n + 6.9 (\log n)^{3/2} + 0.4 \log \log n + 0.6.$$

We obtain a contradiction by using the inequalities (3.4)–(3.8).

3.4 The case of small n

We assume $5\lambda \geq 1.15 \log n$. In this case we have $h = 5\lambda$, and so $A = 24.2$ and $B = 6$, so Theorem 3.1 yields

$$\begin{aligned} \log |\Lambda| &\geq -\frac{2.002 \log x \log(a+1)}{3\lambda} \cdot (24.2)^2 - \frac{2}{3} (2.002 \log x + 3 \log(a+1)) \cdot 2.42 - \\ &\quad - 96 \sqrt{8.008 \log x \log(a+1)} - \log \left(\frac{6.006 \log x \log(a+1)}{\lambda} \cdot 36 \right) - \\ &\quad - \frac{3}{2} \lambda - 10\lambda - 0.15, \end{aligned}$$

and so

$$\log |\Lambda| > -\alpha \log x - \beta \sqrt{\log x} - \log \log x - \gamma, \tag{3.9}$$

where

$$\begin{aligned} \alpha &= \frac{2.002 \log(a+1)}{3\lambda} \cdot (24.2)^2 + \frac{4.004}{3} \cdot 24.2, \\ \beta &= 96 \sqrt{8.008 \log(a+1)}, \\ \gamma &= 48.4 \log(a+1) + \log(216.216) + \log \left(\frac{\log(a+1)}{\lambda} \right) + 11.5\lambda + 0.15. \end{aligned}$$

Since $\log(a+1) < \lambda$, we get

$$\log |\Lambda| > -423.2 \log x - 271.7 \sqrt{\log(a+1)} \sqrt{\log x} - \log \log x - 64.1 \log(a+1) - 5.6.$$

Now using the inequalities $2 \log(a+1) < \log x$ and $\log \log(x+1) < 0.3 \log x$, we get

$$\log |\Lambda| > -648.6 \log x,$$

which combined with (3.2) implies $n < 649$, and so $a \leq 194$. Running a program through the values $167 \leq a \leq 194$ (see `code1.txt` in [30]) and using Lemma 2.7, we conclude that there are no solutions in these cases other than $(1, 1)$.

If $84 \leq a \leq 166$, then by computing the values of α , β and γ , we always get $\alpha < 328.68$, $\beta < 614.59$ and $\gamma < 330.58$. Therefore,

$$n \log x < 328.68 \log x + 614.59 \sqrt{\log x} + \log \log x + 330.58.$$

If $n \geq 557$, the above inequality leads to $n \leq 20909$, and so in this case we have $x < 84n$, which, according to Lemma 2.8, implies that equation (2.1) has a unique positive solution.

In view of Lemma 2.13, we are left with the cases $331 \leq n \leq 547$. By using Lemma 2.6 we find in each case an upper bound for the values of a that have to be considered. Combining (3.2) and (3.9) produces upper bounds for x , but since $y < x$ (see Lemma 2.8), they also serve as upper bounds for y . From this point the proof can be completed using the same method employed in the proof of Lemma 2.11.

Analogous considerations allows one to prove Theorem 1.1 when $2 \leq a \leq 83$.

4 The exceptions of Theorem 1.1

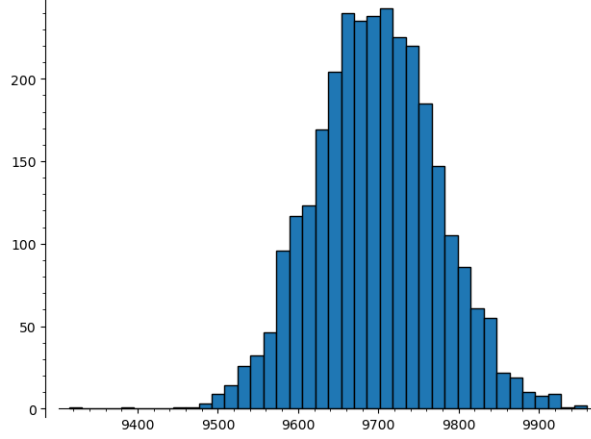
In this chapter we analyze and improve certain steps in Bennett's method from [2] for checking the exceptions of Theorem 1.1.

The first part of Bennett's work consists of checking some "small" cases, namely the ones with $\max\{|x|, |y|\} < 10^{5000}$. This is done in a similar way Lemmas 2.11 and 2.12 were proved. Since solutions correspond to convergents in the continuous fraction expansions of the numbers $\sqrt[n]{1 + 1/a}$, this would come down to inspecting finitely many cases. However, the number of such cases is tremendous and the numbers involved in the computations are huge, so it is clear that one has to reduce them. To do this, the first four potential solutions given by the convergents of the expressions under consideration are tested and it turns out that they do not produce solutions, except for $(1, 1)$. For the remaining ones we always have $q_5 \geq 254$, and so $q_k \geq 254$ for any $k \geq 5$. Inequality (2.3) and Theorem 2.10 imply that a solution to the equation should produce a partial quotient $a_{k+1} > 10^{37}$. On the other hand, by computing the partial quotients under consideration (see `code3.txt` in [30]), it turns out that none of them satisfy this inequality. To be more precise, the following table presents all the partial quotients that exceed 10^6 , denoted by $m(a, n)$.

n	a	$m(a, n)$	n	a	$m(a, n)$	n	a	$m(a, n)$
37	2	1777871	193	21	10564244	271	23	1367976
71	7	2942356	211	39	1642509	277	52	7948681
73	6	1151630	223	22	1599154	281	8	2521773
97	12	2398474	227	8	1747852	281	28	1202167
107	21	2880067	227	23	1844526	283	40	1789726
113	6	17850737	227	42	3682553	293	15	4181758
127	11	3877985	227	57	2130943	293	16	1251851
131	11	8905394	233	67	1085533	293	39	1863042
131	36	2518701	239	42	2589120	293	57	1233501
137	20	1648510	241	64	1225326	317	60	2108163
149	8	2908926	251	61	1009130	317	62	2695313
151	38	2330107	257	4	1815210	331	30	1989372
163	45	2336300	263	76	1125489	337	12	6345557
179	22	1014469	269	41	1114732	337	62	2598313
191	46	18935003	271	18	1330647	347	2	1836676

It should be noted that Bennett did not find the value $m(8, 149) = 2908926$, perhaps due to insufficient precision used. As mentioned in [2], the first 10500 convergents were calculated in each case, which are indeed enough according to our investigation. It turns

out that 10000 would have been enough, as the indexes of the first convergents whose denominators exceed 10^{5000} range between 9315 and 9960. The following histogram illustrates the distribution of these indexes.



The next step consists of using Theorem 7.1 from [2] and combining it with inequality (2.2) to obtain a contradiction, thus proving Theorem 1.2. We quote the statement of this theorem below, but we will only prove it in Chapter 6.

Theorem 4.1. *Let $a < b$ be relatively prime positive integers, and let $n, c(n)$ and $d(n)$ be as in Proposition 6.4. Define $m = \lfloor \frac{n+1}{3} \rfloor$. If*

$$\left(\sqrt[n]{b} - \sqrt[n]{a} \right) e^{c(n)} < 1,$$

then

$$\left| \left(\frac{b}{a} \right)^{1/n} - \frac{p}{q} \right| > \frac{1}{C(n, a, b) q^\lambda},$$

for any positive integers p and q , where

$$C(n, a, b) = 3.15 \cdot 10^{24} (m-1)^2 n^{m-1} e^{c(n)+d(n)} \left(\sqrt[n]{b} + \sqrt[n]{a} \right)^m,$$

$$\lambda = (m-1) \left\{ 1 - \frac{\log \left(\left(\sqrt[n]{b} + \sqrt[n]{a} \right)^m e^{c(n)+1/20} \right)}{\log \left(\left(\sqrt[n]{b} - \sqrt[n]{a} \right)^m e^{c(n)} \right)} \right\}.$$

Combining this result with inequality (2.2) yields

$$y < \left(\frac{C(n, a, a+1)}{na} \right)^{n-\lambda},$$

provided that $\lambda < n$, which turns out to be true in each case.

Considering that $x < 2y$, after running through all the possible cases (see `code2.txt` in [30]), we always get

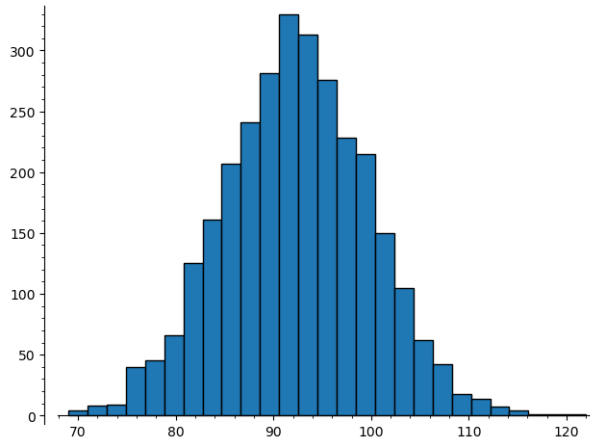
$$\max\{x, y\} < 10^{46},$$

which contradicts the fact that $\max\{x, y\} \geq 10^{5000}$.

As we can see, the bound 10^{5000} is much stronger than what we actually need. As a consequence, this required the program to run through much more cases and much larger numbers than necessary. Of course, such a task requires a tremendous amount of time to be completed; it took about twelve hours for our program to run completely. However, by changing the strategy a bit, we were able to reduce the running time to only a few seconds.

It is better for these two last steps to be done in reverse, that is, first apply Theorem 4.1 to deduce that $\max\{x, y\} < 10^{46}$ and then prove that there are no solutions with $\max\{x, y\} < 10^{46}$ (rather than $\max\{x, y\} < 10^{5000}$).

This significantly reduces the number of convergents to be calculated, as the indexes of the first convergents whose denominator exceed 10^{46} range between 69 and 122 (see their distribution below).



On top of this, we do not need such a strong bound for the partial quotients. None of the partial quotients we found exceeds $2 \cdot 10^7$, so the inequality $a_{k+1} > 10^{37}$ could be relaxed to $a_{k+1} > 2 \cdot 10^7$. For this, we would only need $q_k \geq 3$, so it is enough to consider separately only the case $y = 2$ for each of the values of n and a .

5 Proof of Theorem 2.2 and Theorem 2.3

The proofs of the theorems mentioned in the title are based on the following result derived by Bennet from the proof of [25, Lemma 2.1].

Lemma 5.1. *Let θ be real number and k_0 be a nonnegative integer. Assume that there exist positive real numbers c, d, C and $D > 1$ such that for each positive integer k with $k \geq k_0$, there exist integers p_{lmk} ($0 \leq l, m \leq 1$) with $p_{00k}p_{11k} \neq p_{01k}p_{10k}$, satisfying*

$$|p_{lmk}| \leq c C^k \quad (0 \leq l, m \leq 1)$$

and

$$|p_{l0k} + p_{l1k}\theta| \leq d D^{-k} \quad (0 \leq l \leq 1).$$

Then, if t is any real number with $t > 1$, we may conclude that

$$\left| \theta - \frac{p}{q} \right| > \left(t c C \left(\max \left\{ 1, \frac{td}{t-1} \right\} \right)^{\frac{\log C}{\log D}} \right)^{-1} q^{-1 - \frac{\log C}{\log D}}$$

for all positive integers p and q with $q \geq D^{k_0-1}$.

The proof of this result uses similar ideas to the ones that we will see in the proof of Lemma 6.1, and therefore we do not insist on the details here.

In order to apply this lemma, we first establish some auxiliary results following [5].

5.1 Preliminaries

We define

$$I_l(x) = \frac{1}{2\pi i} \int_{\gamma} \frac{(1+zx)^{k+1/n}}{(z-l)z^k(z-1)^k} dz$$

for $|x| < 1$, where γ is a closed contour with positive orientation that encloses 0 and 1. As shown in [25, Lemma 3.3] and mentioned in [5], we have

$$I_l(x) = p_{l0}(x) + (1+x)^{1/n} p_{l1}(x),$$

for $l \in \{0, 1\}$, where

$$p_{lm}(x) = \sum_{r=0}^{k_{lm}} (-1)^{mr+k_{lm}} \binom{k + \frac{1}{n}}{r} \binom{2k-r-1}{k_{lm}-r} x^r (1+mx)^{k-r},$$

with $k_{lm} = k - 1 + \delta_{lm}$ and δ_{lm} is the Kronecker symbol.

We recall that the generalized binomial $\binom{x}{k}$ with x real and k nonnegative integer is defined by

$$\binom{x}{k} := \frac{x(x-1)(x-2)\dots(x-k+1)}{k!}.$$

Lemma 5.2. *If a and n are positive integers satisfying the hypotheses of either Theorem 2.2 or Theorem 2.3, then*

$$|I_l(1/a)| \leq \frac{5}{na^k(\sqrt{a} + \sqrt{a+1})^{2k}}.$$

Proof. If $n = 3$, we see that the hypothesis implies $(\sqrt{a} + \sqrt{a+1})^2 > \left(\frac{3\sqrt{3}}{2.03}\right)^3$, and so $a \geq 4$. The result then follows directly from Lemma 2.2 of [4], so we can now assume $n \geq 4$.

Following the proof of Lemma 3.2 in [4], we have

$$I_l(1/a) = \frac{|1 - e^{2\pi i/n}|}{2\pi} a^{-2k} \int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1+l/a)(x+1)^k(x+1+1/a)^k} dx. \quad (5.1)$$

By using the inequalities

$$|1 - e^{2\pi i/n}| < \frac{2\pi}{n}$$

and

$$x^{1/n} < x + 1 + l/a,$$

for $x > 0$, we obtain

$$I_l(1/a) < \frac{1}{n} a^{-2k} \int_0^\infty \left(\frac{x}{(x+1)(x+1+1/a)} \right)^k dx.$$

We split the last integral as $\int_0^\infty = \int_0^4 + \int_4^\infty$ and bound from above these last two integrals. For the first one, by considering the function $f : [0, \infty) \rightarrow \mathbb{R}$, $f(x) = \frac{x}{(x+1)(x+1+1/a)}$ and studying its derivative, we easily find

$$\max_{x \in [0, \infty)} f(x) = f\left(\sqrt{\frac{1}{a} + 1}\right) = \frac{a}{(\sqrt{a+1} + \sqrt{a})^2},$$

and consequently

$$\int_0^4 f^k(x) dx \leq \frac{4a^k}{(\sqrt{a+1} + \sqrt{a})^{2k}}.$$

For the second one we have

$$\int_4^\infty f^k(x) dx < \int_4^\infty \left(\frac{x}{(x+1)^2} \right)^k dx < \frac{1}{5^k},$$

provided that $k \geq 4$.

Since $a \geq 3$ (by Remark 2.5), we get

$$\frac{a}{(\sqrt{a+1} + \sqrt{a})^2} > \frac{1}{5},$$

and so

$$\int_0^\infty f^k(x) dx < \frac{5a^k}{(\sqrt{a+1} + \sqrt{a})^{2k}},$$

which finishes the proof for $k \geq 4$.

Let us consider now $k = 1$. We have

$$\int_0^\infty \frac{x^{1+\frac{1}{n}}}{(x+1+l/a)(x+1)(x+1+1/a)} dx < \int_0^\infty \frac{x^{1+1/n}}{(x+1)^3} dx,$$

and with the change of variable $t = \frac{1}{x+1}$, the last integral becomes

$$\int_0^1 t^{-1/n} (1-t)^{1+1/n} dt,$$

which is in fact $B\left(1 - \frac{1}{n}, 2 + \frac{1}{n}\right)$, where B is the Beta function. Using the well-known identity

$$B(z, w) = \frac{\Gamma(z)\Gamma(w)}{\Gamma(z+w)},$$

for $z, w \in \mathbb{C}$ with $\operatorname{Re}(z) > 0$ and $\operatorname{Re}(w) > 0$, we get

$$B\left(1 - \frac{1}{n}, 2 + \frac{1}{n}\right) = \frac{\Gamma\left(1 - \frac{1}{n}\right)\Gamma\left(2 + \frac{1}{n}\right)}{2},$$

and given that Γ is strictly decreasing on $(0, 1]$ and strictly increasing on $[2, \infty)$, the latter expression is bounded above by

$$\frac{\Gamma(0.75)\Gamma(2.25)}{2} < 0.7,$$

and the conclusion follows.

Now, for $k \in \{2, 3\}$, we have

$$\int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1+l/a)(x+1)^k(x+1+1/a)^k} dx < \int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1)^{2k+1}} dx$$

and

$$\begin{aligned} \int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1)^{2k+1}} dx &= \int_0^1 \frac{x^{k+\frac{1}{n}}}{(x+1)^{2k+1}} dx + \int_1^\infty \frac{x^{k+\frac{1}{n}}}{(x+1)^{2k+1}} dx \\ &< \int_0^1 \frac{x^k}{(x+1)^{2k+1}} dx + \int_1^\infty \frac{x^{k+\frac{1}{2}}}{(x+1)^{2k+1}} dx. \end{aligned}$$

The values of the last expression with three exact decimals for $k = 2$ and $k = 3$ are 0.129 and 0.022 respectively, so they are bounded from above by $1/5^{k-1}$, which completes the proof. $\square$

Lemma 5.3. *With the same hypothesis as in Lemma 5.2, we have*

$$|p_{lm}(1/a)| < 1.2 \left(\frac{(\sqrt{a} + \sqrt{a+1})^2}{a} \right)^k.$$

Proof (sketch). This lemma is proved in [4] for $n = 3$. By redoing the calculations from that proof for $n \geq 4$ we get

$$|p_{lm}(1/a)| \leq \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/n} \left(\frac{(\sqrt{a} + \sqrt{a+1})^2}{a} \right)^k$$

We have

$$\sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/n} \leq \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/4}.$$

Moreover, the last expression is decreasing in the variable $t := \sqrt{\frac{a}{a+1}}$ and we saw in Remark 2.5 that $a \geq 3$. This yields $t \geq \sqrt{2/3}$, and so

$$\sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/4} \leq \sqrt{\frac{4}{3}} \left(2 - \sqrt{\frac{3}{4}} \right)^{1/4} < 1.2.$$

□

5.2 Coefficients of $p_{lm}(1/a)$

According to Lemma 4.1 from [7] the number

$$n^j \prod_{p|n} p^{\lfloor \frac{j}{p-1} \rfloor} \binom{k + \frac{1}{n}}{r}$$

is an integer, where n, r, j, k are nonnegative integers with $r \leq j$ and $k \geq 1$. We can therefore define, as in [5],

$$G(n, j, k) = \gcd \left\{ n^j \prod_{p|n} p^{\lfloor \frac{j}{p-1} \rfloor} \binom{k + \frac{1}{n}}{r} \binom{2k - r - 1}{j - r} : r = 0, 1, \dots, j \right\}$$

and

$$G_{n,k} = \gcd\{G(n, k, k), G(n, k-1, k)\}.$$

Lemma 5.4. *Let $k \geq 1$ and $n \geq 3$ be integers. Let p be a prime that does not divide n , and let q be such that $1 \leq q < n/2$ and $pq \equiv \pm 1 \pmod{n}$. If $p \in \left[\frac{k+1}{j}, \frac{nk-n-1}{nj-q} \right]$ for some positive integer $j \leq \sqrt{k/n}$, then p divides $G_{n,k}$.*

Proof (sketch). We start by proving that p does not divide k . Since

$$\frac{nk - n - 1}{nj - q} \leq \frac{nk - n - 1}{n - q} < \frac{nk - n - 1}{n/2} = 2k - \frac{2(n+1)}{n},$$

we get $p < 2k$. Therefore, if p divides k , then $p = k$. For $j = 1$ this is false, because $p \geq k + 1$, and for $j \geq 2$ it is again false because otherwise it would imply

$$2n - q \leq nj - q \leq n - \frac{n+1}{k},$$

and so $n + \frac{n+1}{k} \leq q < \frac{n}{2}$, a contradiction.

Now we establish the inequality $p \geq \sqrt{nk+1}$. We indeed have

$$p \geq \frac{k+1}{j} \geq \frac{k+1}{\sqrt{k/n}} = \sqrt{nk} + \sqrt{\frac{n}{k}} > \sqrt{nk+1}.$$

Finally, let us notice that

$$p \leq \frac{nk - n - 1}{nj - q} < \frac{n(k-1)}{nj - q},$$

and so

$$\frac{k-1}{p} \geq \frac{nj - q}{n} = (j-1) + \frac{n-q}{n}.$$

Moreover, $j > \frac{k-1}{p}$ and, since $\frac{n-q}{n} \in (0, 1)$, we get

$$\left\{ \frac{k-1}{p} \right\} > \frac{n-q}{n}.$$

With all these ingredients, the proof can be completed following the lines of the proof of [4, Lemma 3.1]. $\square$

In the proof of the next lemma, we use the following notations:

$$\begin{aligned} \theta(x) &= \sum_{p \leq x} \log p, \\ \theta(x; k, l) &= \sum_{\substack{p \leq x \\ p \equiv l \pmod{k}}} \log p, \end{aligned}$$

where p symbolizes a prime variable.

Lemma 5.5. *For $n \neq 4$ and c_n appearing in Table 1 and any positive integer k , the following inequality holds*

$$G_{n,k} > e^{-16} c_n^k.$$

For $n = 4$,

$$G_{4,k} > e^{-20} c_3^k.$$

Proof outline. The strategy of the proof is to lower bound $G_{n,k}$ by the product of primes described in Lemma 5.4 and then to use estimates for products of primes in intervals to lower bound these products. This will prove the inequalities for all values of k , with only a few exceptions for which they are checked directly by the computer.

Let us notice that the range of q in the Lemma 5.4 covers all the possible residues of p modulo n . Moreover, each of the intervals

$$\left[\frac{k+1}{j}, \frac{nk-n-1}{nj-q} \right]$$

is contained in

$$\left[\frac{k+1}{j}, \frac{nk-n-1}{nj-1} \right],$$

so for any prime p in this last interval (regardless of its residue modulo n) we have $p \mid G_{n,k}$. We will treat these primes all together and then account for the extra primes given by Lemma 5.4, i.e. for each q with $1 < q < n/2$ we will consider the primes in the interval

$$\left(\frac{nk-n-1}{nj-1}, \frac{nk-n-1}{nj-q} \right]$$

that are congruent to $\pm q^{-1}$ modulo n .

We will present the computations for $n = 7$. For any other value of n the proof follows the same lines.

Let us consider the sets

$$\begin{aligned} I_1 &= \bigcup_{j=1}^{\lfloor \sqrt{k/7} \rfloor} \left[\frac{k+1}{j}, \frac{7k-8}{7j-1} \right], \\ I_2 &= \bigcup_{j=1}^{\lfloor \sqrt{k/7} \rfloor} \left(\frac{7k-8}{7j-1}, \frac{7k-8}{7j-2} \right], \\ I_3 &= \bigcup_{j=1}^{\lfloor \sqrt{k/7} \rfloor} \left(\frac{7k-8}{7j-1}, \frac{7k-8}{7j-3} \right]. \end{aligned}$$

We first assume that $10^6 \leq k < 10^{11}$. Using the inequality $\theta(x) < x$ for $x \leq 10^{11}$ (see [24]), we get

$$\begin{aligned} \sum_{p \in \left[\frac{k+1}{j}, \frac{7k-8}{7j-1} \right]} \log p &\geq \theta \left(\frac{7k-8}{7j-1} \right) - \theta \left(\frac{k+1}{j} \right) \\ &> \theta \left(\frac{7k-8}{7j-1} \right) - \frac{k+1}{j}. \end{aligned}$$

For each $1 \leq j \leq 19$ we use Theorem C to obtain inequalities of the form

$$\begin{aligned} \theta\left(\frac{7k-8}{7j-1}\right) &> \frac{7k-8}{7j-1} \left(1 - \frac{1}{c_j \ln \frac{7k-8}{7j-1}}\right) \\ &\geq \frac{7k-8}{7j-1} \left(1 - \frac{1}{c_j \ln \frac{7 \cdot 10^6 - 8}{7j-1}}\right). \end{aligned}$$

After summing all these inequalities, we get

$$\sum_{p \in I_1} \log p \geq 0.24299k - 8.27977.$$

For the primes in I_2 and I_3 we will use bounds for products of primes in arithmetic progressions from [24] (see the appendix), thus getting

$$\begin{aligned} \sum_{\substack{p \in I_2 \\ p \equiv 3 \pmod{7}}} \log p &\geq \sum_{j=1}^{19} \left(\theta\left(\frac{7k-8}{7j-2}; 7, 3\right) - \theta\left(\frac{7k-8}{7j-1}; 7, 3\right) \right) \\ &\geq \sum_{j=1}^{19} \left(\frac{1}{6} \cdot \frac{7k-8}{7j-2} - 0.829249 \sqrt{\frac{7k-8}{7j-2}} - \right. \\ &\quad \left. - \frac{1}{6} \cdot \frac{7k-8}{7j-1} - 0.829249 \sqrt{\frac{7k-8}{7j-1}} \right) \\ &> 0.05539k - 4.78573\sqrt{7k-8} - 0.06332 \\ &> 0.04272k - 0.06332, \end{aligned}$$

$$\begin{aligned} \sum_{\substack{p \in I_2 \\ p \equiv 4 \pmod{7}}} \log p &\geq \sum_{j=1}^{19} \left(\theta\left(\frac{7k-8}{7j-2}; 7, 4\right) - \theta\left(\frac{7k-8}{7j-1}; 7, 4\right) \right) \\ &\geq \sum_{j=1}^{19} \left(\frac{1}{6} \cdot \frac{7k-8}{7j-2} - \left(\frac{1}{3} + 0.829249\right) \sqrt{\frac{7k-8}{7j-2}} - \right. \\ &\quad \left. - \frac{1}{6} \cdot \frac{7k-8}{7j-1} + \left(\frac{1}{3} - 0.829249\right) \sqrt{\frac{7k-8}{7j-1}} \right) \\ &> 0.05539k - 4.81023\sqrt{7k-8} - 0.06332 \\ &> 0.04266k - 0.06332, \end{aligned}$$

$$\begin{aligned}
\sum_{\substack{p \in I_3 \\ p \equiv 2 \pmod{7}}} \log p &\geq \sum_{j=1}^{19} \left(\theta\left(\frac{7k-8}{7j-3}; 7, 2\right) - \theta\left(\frac{7k-8}{7j-1}; 7, 2\right) \right) \\
&\geq \sum_{j=1}^{19} \left(\frac{1}{6} \cdot \frac{7k-8}{7j-3} - \left(\frac{1}{3} + 0.829249 \right) \sqrt{\frac{7k-8}{7j-3}} - \right. \\
&\quad \left. - \frac{1}{6} \cdot \frac{7k-8}{7j-1} + \left(\frac{1}{3} - 0.829249 \right) \sqrt{\frac{7k-8}{7j-1}} \right) \\
&> 0.13225k - 4.91479\sqrt{7k-8} - 0.15115 \\
&> 0.11924k - 0.15115
\end{aligned}$$

and

$$\begin{aligned}
\sum_{\substack{p \in I_3 \\ p \equiv 5 \pmod{7}}} \log p &\geq \sum_{j=1}^{19} \left(\theta\left(\frac{7k-8}{7j-3}; 7, 5\right) - \theta\left(\frac{7k-8}{7j-1}; 7, 5\right) \right) \\
&\geq \sum_{j=1}^{19} \left(\frac{1}{6} \cdot \frac{7k-8}{7j-3} - 0.829249 \sqrt{\frac{7k-8}{7j-3}} - \right. \\
&\quad \left. - \frac{1}{6} \cdot \frac{7k-8}{7j-1} - 0.829249 \sqrt{\frac{7k-8}{7j-1}} \right) \\
&> 0.13225k - 4.86030\sqrt{7k-8} - 0.15115 \\
&> 0.11939k - 0.15115.
\end{aligned}$$

Finally, after summing the above inequalities, we get

$$\sum_{p \in I_1 \cup I_2 \cup I_3} \log p > 0.567k - 8.70871 > \log(1.76)k = c_7k,$$

and so

$$\log G_{7,k} > c_7k,$$

as desired.

The case $k \geq 10^{11}$ is similar. The only differences are that this time we use the inequality $\theta(x) < 1.000081x$ (true for any $x > 0$) instead of $\theta(x) < x$, and we estimate the expressions $\theta(x; k, l)$ using Theorem B.

The case $k < 10^6$ is checked with the computer in the following way: for each value of k the product of primes described in Lemma 5.4 is computed and it is verified if it is greater than $e^{-20}c_3^k$ if $n = 3$, or greater than $e^{-16}c_n^k$ if $n \neq 3$. This is true for all the values of k , except for a couple of exceptions. For each exception the inequality is proved by directly computing $G_{n,k}$ and comparing it with the value of the right hand side. $\square$

5.3 Proof of Theorem 2.2

We define for a and n satisfying the conditions of either Theorem 2.2 or 2.3 and each $k > 0$ and $l, m \in \{0, 1\}$ the number

$$p_{lmk} = (na)^k p_{lm}(1/a) G_{n,k}^{-1} \prod_{p|n} p^{\lfloor \frac{k}{p-1} \rfloor}.$$

Notice that p_{lmk} is an integer, as a consequence of the result from the beginning of the previous section.

We intend to apply Lemma 5.1 for these numbers, so we want to make sure the expression

$$p_{00k}p_{11k} - p_{01k}p_{10k}$$

is nonzero. This comes down to checking that the determinant

$$\begin{vmatrix} p_{00}(1/a) & p_{01}(1/a) \\ p_{10}(1/a) & p_{11}(1/a) \end{vmatrix}$$

is nonzero. A formula for calculating determinants of this form is provided in [25, Lemma 3.4] and it immediately follows from it that the determinant above is indeed nonzero.

We are now going to derive numbers c, d, C, D satisfying the hypotheses of Lemma 5.1.

From Lemma 5.3 and Lemma 5.5 we have

$$|p_{lmk}| \leq 1.2e^{16} n^k c_n^{-k} u_n^k (\sqrt{a} + \sqrt{a+1})^{2k},$$

so we can choose

$$\begin{aligned} c &= 1.2e^{16}, \\ C &= \frac{nu_n}{c_n} (\sqrt{a} + \sqrt{a+1})^2. \end{aligned}$$

From Lemma 5.2 and Lemma 5.5 we have

$$\begin{aligned} |p_{l0k} + p_{l1k}\theta| &= (na)^k G_{n,k}^{-1} \prod_{p|n} p^{\lfloor \frac{k}{p-1} \rfloor} |p_{l0}(1/a) + p_{l1}(1/a)\theta| \\ &\leq \frac{5}{n} e^{16} n^k c_n^{-k} u_n^k \cdot \frac{1}{(\sqrt{a} + \sqrt{a+1})^{2k}}, \end{aligned}$$

so we can choose

$$\begin{aligned} d &= \frac{5}{n} e^{16}, \\ D &= \frac{c_n}{nu_n} (\sqrt{a} + \sqrt{a+1})^2. \end{aligned}$$

Taking $t = 3$ and $k_0 = 1$ yields

$$\left| \theta - \frac{p}{q} \right| > \left(\frac{0.48n^2 u_n (\sqrt{a} + \sqrt{a+1})^2}{c_n} \right)^{-1} \left(\frac{7.5e^{16}}{n} q \right)^{-\lambda_1},$$

so, in order to prove Theorem 2.2, it would be enough to prove the inequality

$$\left(\frac{10^{10}n}{7.5e^{16}} \right)^{\lambda_1} > \frac{0.48n^2 u_n}{c_n} \left(1 + \sqrt{1 + \frac{1}{a}} \right)^2.$$

We easily check this with a program (see `code5.txt` in [30]) after using the inequalities $\lambda_1 > 2$ and $a \geq 3$, thus completing the proof of Theorem 2.2.

5.4 Proof of Theorem 2.3

To prove Theorem 2.3 we will use Lemma 5.1 with the same choices for p_{lmk} as before. In order to find suitable values for c, d, C and D , we will need to strengthen a bit some of the previous results.

Instead of using Lemma 5.3, we will use the inequality

$$|p_{lm}(1/a)| < \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/n} \left(\frac{(\sqrt{a} + \sqrt{a+1})^2}{a} \right)^k,$$

which was actually a step in the proof of Lemma 5.3. Therefore, the choices for c and C are

$$c = \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/n},$$

$$C = nu_n(\sqrt{a} + \sqrt{a+1})^2.$$

Bounding $|p_{l0k} + p_{l1k}\theta|$ as in the previous section, except that $G_{n,k}$ is bounded from below by 1 this time, yields the following choices:

$$d = \frac{5}{n},$$

$$D = \frac{1}{nu_n}(\sqrt{a} + \sqrt{a+1})^2.$$

Assuming that $n \geq 14$ and choosing t such that $\frac{td}{t-1} = 1$, that is, $t = \frac{n}{n-5}$, and $k_0 = 1$ in Lemma 5.1, gives us the inequality

$$\left| \theta - \frac{p}{q} \right| > \left(\frac{n^2}{n-5} \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{1/n} u_n(\sqrt{a} + \sqrt{a+1})^2 \right)^{-1} q^{-\lambda_2},$$

so in order to prove Theorem 2.3, it would be enough to show that

$$\frac{8(n-5)}{n} > \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}}\right)^{1/n} \left(1 + \sqrt{\frac{a+1}{a}}\right)^2.$$

This inequality is indeed true, as it can be shown that the left hand side is bounded from below by 5, while the right hand side is bounded from above by 5. Notice that to achieve this we need to prove that $a \geq 5$, which can be justified as in the proof of Remark 2.5.

Now let's consider the case $n \leq 13$.

Using the equality (5.1), we obtain

$$|I_l(1/a)| \leq \frac{|1 - e^{2\pi i/n}|}{2\pi} a^{-2k} \int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1)^{k+1}(x+1+1/a)^k} dx. \quad (5.2)$$

This further implies

$$|I_l(1/a)| \leq \frac{|1 - e^{2\pi i/n}|}{2\pi} a^{-2k} \int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1)^{2k+1}} dx.$$

Now, in order to improve Lemma 5.2, we would need to bound from above the expression

$$\frac{|1 - e^{2\pi i/n}|}{2\pi} \left(\frac{(\sqrt{a} + \sqrt{a+1})^2}{a} \right)^k \int_0^\infty \frac{x^{k+\frac{1}{n}}}{(x+1)^{2k+1}} dx. \quad (5.3)$$

Let us notice that

$$|1 - e^{2\pi i/n}| = 2 \sin(\pi/n)$$

and

$$\int_0^\infty \frac{x^{k+1/n}}{(x+1)^{2k+1}} dx = B\left(k + \frac{1}{n} + 1, k - \frac{1}{n}\right).$$

Moreover, since the second factor in (5.3) is decreasing in a , the expression in (5.3) is bounded from above by

$$\frac{\sin(\pi/n)}{\pi} \left(\frac{(\sqrt{a_n} + \sqrt{a_n+1})^2}{a_n} \right)^k B\left(k + \frac{1}{n} + 1, k - \frac{1}{n}\right),$$

where a_n is the smallest value of a satisfying the hypothesis of Theorem 2.3 for a given n . Computing the above expression for all the possible cases with $k < 26$, we get

$$|I_l(1/a)| \leq \frac{d_n}{a^k (\sqrt{a} + \sqrt{a+1})^{2k}}, \quad (5.4)$$

where $d_3 = 0.475$ and $d_n = 0.374$ for $4 \leq n \leq 13$.

Let us consider now the case $k \geq 26$. By studying the derivative of $\frac{x^{1/n}}{x+1}$ one easily sees that it is positive on the interval $(0, \frac{1}{n-1})$ and negative on the interval $(\frac{1}{n-1}, \infty)$, and so $\frac{1}{n-1}$ is a maximum point for $\frac{x^{1/n}}{x+1}$ (considered over $[0, \infty)$). This implies

$$\frac{x^{\frac{1}{n}}}{x+1} \leq \frac{(n-1)^{\frac{n-1}{n}}}{n},$$

for any $x \geq 0$. Using this inequality in (5.2) yields

$$|I_l(1/a)| \leq \frac{|1 - e^{2\pi i/n}|}{2\pi} a^{-2k} \cdot \frac{(n-1)^{\frac{n-1}{n}}}{n} J_k(a), \quad (5.5)$$

where

$$J_k(a) := \int_0^\infty \left(\frac{x}{(x+1)(x+1+1/a)} \right)^k dx.$$

Now we split this last integral as $\int_0^{1/2} + \int_{1/2}^2 + \int_2^\infty$ and bound each term from above. For the first one we use the inequality $\frac{x}{(x+1)^2} \leq 2/9$ for $x \in [0, 1/2]$. For the second one we proceed exactly as we did in the proof of Lemma 5.2, thus getting

$$\int_{1/2}^2 \left(\frac{x}{(x+1)(x+1+1/a)} \right)^k dx \leq \frac{3}{2} \cdot \frac{a^k}{(\sqrt{a} + \sqrt{a+1})^{2k}},$$

and we bound the third one by the series $\sum_{N=2}^\infty \left(\frac{N}{(N+1)^2} \right)^k$. Putting these together, we get

$$J_k(a) \leq \frac{1}{2} \cdot \left(\frac{2}{9} \right)^k + \frac{3}{2} \cdot \frac{a^k}{(\sqrt{a} + \sqrt{a+1})^{2k}} + \sum_{N=2}^\infty \left(\frac{N}{(N+1)^2} \right)^k.$$

For each of the values of n under consideration the hypothesis implies $a \geq 6$, and so

$$\frac{(\sqrt{a} + \sqrt{a+1})^2}{a} < 4.327. \quad (5.6)$$

Therefore,

$$\left(\frac{2}{9} \right)^k < \frac{a^k}{(\sqrt{a} + \sqrt{a+1})^{2k}}.$$

Finally,

$$\begin{aligned}
\frac{(\sqrt{a} + \sqrt{a+1})^{2k}}{a^k} \sum_{N=2}^{\infty} \left(\frac{N}{(N+1)^2} \right)^k &< \sum_{N=2}^{\infty} \left(\frac{4.327N}{(N+1)^2} \right)^k \\
&\leq \sum_{N=2}^{\infty} \left(\frac{4.327N}{(N+1)^2} \right)^{26} \\
&< \sum_{N=2}^3 \left(\frac{4.327N}{(N+1)^2} \right)^{26} + \sum_{N=4}^{\infty} \frac{(4.237)^{26}}{(N+1)^{26}} \\
&< 0.37 + (4.237)^{26} \left(\zeta(26) - \sum_{N=1}^4 \frac{1}{N^{26}} \right) \\
&< 0.5.
\end{aligned}$$

Putting all these pieces together, we obtain

$$J_k(a) \leq 2.5 \cdot \frac{a^k}{(\sqrt{a} + \sqrt{a+1})^{2k}}. \quad (5.7)$$

Computing the values of the expression

$$\frac{|1 - e^{2\pi i/n}|}{2\pi} \cdot \frac{(n-1)^{\frac{n-1}{n}}}{n},$$

we find that it is always bounded above by 0.146. Using this and (5.7) in (5.5) yields (5.4) for $k \geq 26$. It therefore follows that we can choose $d = d_n$ in Lemma 5.1 and D as in the beginning of the section. Reusing the old c and C , taking $k_0 = 2$ and $t = 1.598$ for $4 \leq n \leq 13$ yields the result of Theorem 2.3 for any q with

$$q \geq \frac{(\sqrt{a} + \sqrt{a+1})^2}{nu_n}.$$

To see this, we just need to prove that

$$1.598 \sqrt{\frac{a+1}{a}} \left(2 - \sqrt{\frac{a}{a+1}} \right)^{\frac{1}{n}} \left(1 + \sqrt{\frac{a+1}{a}} \right)^2 < 8.$$

The above expression is clearly decreasing in the variable $s = \sqrt{\frac{a}{a+1}}$. Using this fact and the inequalities $s \geq \sqrt{5/6}$ and $n \geq 4$ proves the inequality.

For $n = 3$ we take $t = 1.905$ and reach the same conclusion.

To finish the proof, it remains to prove the inequality in Theorem 2.3 for $3 \leq n \leq 13$ and q satisfying

$$q < \frac{(\sqrt{a} + \sqrt{a+1})^2}{nu_n}.$$

By the mean value theorem, we have

$$\left| \sqrt[n]{1 + \frac{1}{a}} - 1 \right| = \frac{1}{na\xi^{\frac{n-1}{n}}},$$

with $\xi \in (1, 1 + \frac{1}{a})$, and so

$$\left| \sqrt[n]{1 + \frac{1}{a}} - 1 \right| > \frac{1}{na \left(1 + \frac{1}{a}\right)^{\frac{n-1}{n}}} > \frac{1}{8na} > \frac{1}{8nu_n a q^{\lambda_2}}.$$

This proves the theorem in the case $p \leq q$, because

$$\left| \sqrt[n]{1 + \frac{1}{a}} - \frac{p}{q} \right| \geq \left| \sqrt[n]{1 + \frac{1}{a}} - 1 \right|.$$

Now, if $p \geq q + 1$, then

$$\frac{p}{q} \geq 1 + \frac{1}{q} > 1 + \frac{nu_n}{(\sqrt{a} + \sqrt{a+1})^2} > 1 + \frac{n}{4.237a},$$

where the last inequality follows from (5.6) and $u_n \geq 1$. Using the inequality

$$\sqrt[n]{1 + \frac{1}{a}} \leq 1 + \frac{1}{na}, \quad (5.8)$$

we get

$$\left| \sqrt[n]{1 + \frac{1}{a}} - \frac{p}{q} \right| \geq \frac{p}{q} - \sqrt[n]{1 + \frac{1}{a}} \geq \frac{n}{4.237a} - \frac{1}{na},$$

and we can easily verify that

$$\frac{n}{4.237a} - \frac{1}{na} > \frac{1}{8na},$$

thus completing the proof as in the previous case.

Theorem 2.3 is now proved.

Remark. Inequality (5.8) is obvious after being raised to the power n . In order to derive it, one can use the Maclaurin series of the function $\sqrt{x+1}$, or can apply the inequality between the arithmetic mean and geometric mean of the numbers $1 + \frac{1}{a}, 1, 1, \dots, 1$ (there are $n-1$ copies of 1).

6 Padé approximation

The goal of this chapter is to provide a sketch of the proof of Theorem 4.1. For this we will use a Liouville-type result established in [2, Lemma 2.1]. We state and prove this result in section 6.1 and then in section 6.2 we provide some considerations on Padé approximation that we will need in order to apply the result. Combining all these tools, we will sketch the proof of Theorem 4.1 in section 6.3.

6.1 Auxiliary lemma

Lemma 6.1 ([2], Lemma 2.1). *Let θ, c, d, C and D be positive real numbers and r_0 and k positive integers, with θ irrational and $C, D > 1$. Suppose that for each positive integer r with $r \geq r_0$, there is a sequence of polynomials*

$$P_{i,r}(x) = \sum_{j=0}^k a_{ij} x^j$$

for $0 \leq i \leq k$, with $a_{ij} \in \mathbb{Z}$, such that the matrix $A = (a_{ij})_{0 \leq i, j \leq k}$ is nonsingular,

$$|a_{ij}| \leq cC^r$$

and

$$|P_{i,r}(\theta)| \leq dD^{-r}.$$

If t is any real number satisfying $t > 1$ and $tdD^{r_0-1} \geq 1$, p and q are positive integers with $tdq^k \geq D^{r_0-1}$ and $\delta = \max\{|\theta|, |p/q|, 1\}$, then

$$\left| \theta - \frac{p}{q} \right| > \left(\frac{t}{t-1} \cdot \frac{k(k+1)}{2} \cdot \delta^{k-1} \cdot c \cdot C \cdot (td)^{\frac{\log C}{\log D}} \right)^{-1} q^{-\lambda},$$

where

$$\lambda = k \left(1 + \frac{\log C}{\log D} \right).$$

Proof. Let r be the smallest positive integer that satisfies $D^r > tdq^k$. Clearly $r \geq r_0$ and, by the minimality of r , we have $D^{r-1} \leq tdq^k$. This further implies

$$C^r = (D^r)^{\frac{\log C}{\log D}} \leq (Dtdq^k)^{\frac{\log C}{\log D}} = C \cdot (td)^{\frac{\log C}{\log D}} \cdot q^{\frac{k \log C}{\log D}}. \quad (6.1)$$

Since the matrix A is nonsingular, there exists some $0 \leq i \leq k$ such that $P_{i,r}(p/q) \neq 0$. Indeed, otherwise we have $P_{i,r}(p/q) = 0$ for every $0 \leq i \leq k$, which produces the matrix equality

$$A \cdot \begin{pmatrix} 1 & 1 & \dots & 1 \\ p/q & p/q & \dots & p/q \\ (p/q)^2 & (p/q)^2 & \dots & (p/q)^2 \\ \dots & \dots & \dots & \dots \\ (p/q)^k & (p/q)^k & \dots & (p/q)^k \end{pmatrix} = 0.$$

Since A is nonsingular, this equality should imply that the second matrix is the zero matrix, which is clearly a contradiction.

Now, $q^k |P_{i,r}(p/q)|$ is a positive integer and so $q^k |P_{i,r}(p/q)| \geq 1$. Thus

$$\left| P_{i,r} \left(\frac{p}{q} \right) \right| \geq \frac{1}{q^k}.$$

On the other hand,

$$\begin{aligned} \left| P_{i,r} \left(\frac{p}{q} \right) \right| &\leq \left| P_{i,r} \left(\frac{p}{q} \right) - P_{i,r}(\theta) \right| + |P_{i,r}(\theta)| \\ &< \left| P_{i,r} \left(\frac{p}{q} \right) - P_{i,r}(\theta) \right| + \frac{1}{tq^k}. \end{aligned}$$

Combining the last two inequalities yields

$$\left| P_{i,r} \left(\frac{p}{q} \right) - P_{i,r}(\theta) \right| > \left(1 - \frac{1}{t} \right) q^{-k}. \quad (6.2)$$

Finally,

$$\begin{aligned} \left| P_{i,r} \left(\frac{p}{q} \right) - P_{i,r}(\theta) \right| &= \left| \int_{p/q}^{\theta} P'_{i,r}(x) dx \right| \\ &\leq \left| \int_{p/q}^{\theta} |P'_{i,r}(x)| dx \right| \\ &\leq \left| \int_{p/q}^{\theta} (|a_{i1}| + 2|a_{i2}||x| + 3|a_{i3}||x|^2 + \dots + k|a_{ik}||x|^{k-1}) dx \right| \\ &\leq \left| \int_{p/q}^{\theta} (1 + 2 + 3 + \dots + k) cC^r \delta^{k-1} dx \right| \\ &= \frac{k(k+1)}{2} cC^r \delta^{k-1} \left| \theta - \frac{p}{q} \right|. \end{aligned}$$

The conclusion follows using this last inequality along with (6.2) and (6.1). $\square$

To find suitable choices for the polynomials in the hypothesis of the above lemma, we will use Padé approximants to functions of the form $(1 - z)^k$ as in the next section.

6.2 Padé approximants to binomial functions

Let $f_1, \dots, f_m$ be analytic functions at 0. Let $n_1, \dots, n_m$ be nonnegative integers and let $P_1, \dots, P_m$ be polynomials in x of degrees $n_1, \dots, n_m$, respectively, such that the function

$$R(x) := P_1(x)f_1(x) + \dots + P_m(x)f_m(x)$$

has a zero at $x = 0$ of order at least equal to $\sum_{k=1}^m (n_k + 1) - 1$.

The function R is called a remainder function in the Padé approximation to the functions $f_1, \dots, f_m$ at 0 with weights $n_1, \dots, n_m$. The polynomials $P_1, \dots, P_m$ are called Padé approximants with weights $n_1, \dots, n_m$.

We define

$$\phi(\zeta) = \prod_{k=1}^m \prod_{h=0}^{\rho_k-1} (\zeta - \omega_k - h)$$

and

$$R\left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array} \right. \right) = \frac{(-1)^{\sigma-1} \Gamma(\rho_1) \cdots \Gamma(\rho_m)}{2\pi i} \int_{\gamma} \frac{(1-x)^{\zeta}}{\phi(\zeta)} d\zeta,$$

where γ is a closed contour with positive orientation that encloses all the poles of the integrand, $\rho_1, \dots, \rho_m$ are positive integers, $\omega_1, \dots, \omega_m$ are complex numbers such that $\omega_i - \omega_j$ are non-integral for all $i \neq j$, and $\sigma = \rho_1 + \dots + \rho_m$.

Using the Cauchy residue formula, the following identity is established in [19]:

$$R\left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array} \right. \right) = \sum_{j=1}^m A_j \left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array} \right. \right) (1-x)^{\omega_j},$$

where

$$A_j \left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array} \right. \right) = (-1)^{\sigma-1} \Gamma(\rho_1) \cdots \Gamma(\rho_m) \sum_{k=0}^{\rho_j-1} \frac{(1-x)^k}{\phi'(\omega_j + k)}.$$

It easily follows from here that A_j is a polynomial in x of degree $\rho_j - 1$, for every $j \in \{1, \dots, m\}$. Moreover, as proved in [19], R has a zero of order $\sigma - 1$ at $x = 0$, so R is the remainder function in the Padé approximation to the functions $(1-x)^{\omega_j}$ and the polynomials A_j constitute a system of Padé approximants to the previously mentioned functions.

We define

$$R_i \left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array} \right. \right) = R \left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 + \delta_{i1} & \cdots & \rho_m + \delta_{im} \end{array} \right. \right)$$

and

$$A_{ij} \left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array} \right. \right) = A_j \left(x \left| \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 + \delta_{i1} & \cdots & \rho_m + \delta_{im} \end{array} \right. \right),$$

where δ_{ij} is the Kronecker symbol.

Let us consider the weights $\omega_1, \dots, \omega_m$ fixed for now and define

$$R_i(x, r) = R_i\left(x \mid \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array}\right)$$

and

$$A_{ij}(x, r) = A_{ij}\left(x \mid \begin{array}{ccc} \omega_1 & \cdots & \omega_m \\ \rho_1 & \cdots & \rho_m \end{array}\right).$$

We establish some bounds for these expressions.

Lemma 6.2 ([2], Lemma 3.1). *Let r, i and m be positive integers. Then*

$$|R_i(x, r)| \leq \frac{|x|^m}{(m-1)!} |1 - (1-x)^{1/m}|^{mr},$$

for any $x < 0$.

Proof. As in [19], we have

$$R_i(x, r) = \int_0^x \int_0^{t_1} \cdots \int_0^{t_{m-2}} R_i(x \mid t_1, \dots, t_{m-1}) dt_{m-1} \cdots dt_1, \quad (6.3)$$

for

$$x \leq t_1 \leq \dots \leq t_{m-1} \leq 0,$$

where

$$R_i(x \mid t_1, \dots, t_{m-1}) = \prod_{k=1}^m \left(\frac{t_{k-1} - t_k}{1 - t_{k-1}} \right)^{r + \delta_{ik}} \prod_{k=2}^m (1 - t_{k-1})^{\frac{1}{n} - 1},$$

with $t_0 = x$ and $t_m = 0$.

From here we get

$$|R_i(x \mid t_1, \dots, t_{m-1})| \leq \left| \frac{t_{i-1} - t_i}{1 - t_{i-1}} \right| \cdot \prod_{k=1}^m \left(\frac{t_k - t_{k-1}}{1 - t_{k-1}} \right)^r.$$

Now

$$\left| \frac{t_{i-1} - t_i}{1 - t_{i-1}} \right| = \frac{t_i - t_{i-1}}{1 - t_{i-1}} \leq \frac{t_{i-1}^2 - t_{i-1}}{1 - t_{i-1}} = -t_{i-1} \leq -x,$$

whence

$$\left| \frac{t_{i-1} - t_i}{1 - t_{i-1}} \right| \leq |x|,$$

and we can bound $\prod_{k=1}^m \left(\frac{t_k - t_{k-1}}{1 - t_{k-1}} \right)$ using the argument from the proof of Corollary 2.3 of [7], namely that the only critical point of this expression is the one given by

$$t_k = 1 - (1-x)^{1-\frac{k}{m}},$$

for $k = \overline{1, m-1}$.

We conclude that

$$|R_i(x \mid t_1, \dots, t_{m-1})| \leq |x| \left| 1 - (1-x)^{1/m} \right|^{mr},$$

and using this in (6.3) yields the result. $\square$

Lemma 6.3 ([2], Lemma 3.2). *Let $n > m \geq 2$ and $r \geq 1$ be integers and*

$$\Phi_{m,n,r} = \max \left\{ \frac{n^2}{n-1} r^{1/n}, \frac{n^2}{(m-1)(n-m+1)} r^{(m-1)/n} \right\}.$$

Then

$$|A_{ij}(x, r)| \leq \left(1 + \max \left\{ 1, \frac{1-x}{6} \right\} \right) (r+1) \Phi_{m,n,r}^{m-1} (1 + (1-x)^{1/m})^{mr},$$

for any $x < 0$.

We define $\Delta_{m,n,r}$ to be the smallest positive rational number for which $\Delta_{m,n,r} A_{ij}(x, r)$ has integer coefficients. The following result is proved in [2, Proposition 6.1].

Proposition 6.4. *For any prime $17 \leq n \leq 347$ and $m = \lfloor \frac{n+1}{3} \rfloor$, the following inequality holds:*

$$\log \Delta_{m,n,r} < c(n)r + d(n),$$

where the constants $c(n)$ and $d(n)$ are defined below.

n	$c(n)$	$d(n)$	n	$c(n)$	$d(n)$	n	$c(n)$	$d(n)$
17	8.93	13.06	107	83.55	50.84	227	201.15	116.91
19	9.40	15.46	109	84.18	58.97	229	202.11	100.61
23	13.03	17.66	113	89.22	77.93	233	207.50	102.49
29	17.39	29.95	127	100.47	72.61	239	213.74	105.66
31	17.92	30.55	131	105.34	71.51	241	214.95	95.14
37	21.92	32.51	137	111.14	79.94	251	226.83	115.64
41	25.83	36.08	139	112.15	77.27	257	233.75	113.23
43	26.62	33.95	149	122.53	85.82	263	240.15	119.49
47	30.46	45.38	151	123.41	89.04	269	246.54	124.75
53	34.78	35.37	157	129.07	81.61	271	247.72	134.21
59	39.18	48.34	163	134.80	93.64	277	254.62	119.17
61	39.96	55.93	167	139.95	82.87	281	260.46	116.79
67	44.76	43.56	173	146.07	87.71	283	261.67	118.21
71	48.36	54.80	179	151.40	83.92	293	274.23	129.73
73	52.83	48.11	181	152.20	91.69	307	289.00	124.89
79	58.27	54.65	191	163.78	84.40	311	294.70	130.14
83	62.70	49.64	193	164.81	91.51	313	296.38	130.18
89	67.56	60.29	197	170.17	104.53	317	302.73	143.63
97	73.71	62.14	199	170.80	110.41	321	317.41	147.69
101	78.29	50.36	211	183.12	124.02	327	324.63	139.95
103	79.16	60.85	223	195.74	112.93	347	338.02	133.98

6.3 Main result

For convenience, we state Theorem 4.1 again.

Theorem. *Let $a < b$ be relatively prime positive integers, and let $n, c(n)$ and $d(n)$ be as in Proposition 6.4. Define $m = \lfloor \frac{n+1}{3} \rfloor$. If*

$$\left(\sqrt[m]{b} - \sqrt[m]{a} \right) e^{c(n)} < 1,$$

then

$$\left| \left(\frac{b}{a} \right)^{1/n} - \frac{p}{q} \right| > \frac{1}{C(n, a, b) q^\lambda},$$

for any positive integers p and q , where

$$C(n, a, b) = 3.15 \cdot 10^{24} (m-1)^2 n^{m-1} e^{c(n)+d(n)} \left(\sqrt[m]{b} + \sqrt[m]{a} \right)^m,$$

$$\lambda = (m-1) \left\{ 1 - \frac{\log \left(\left(\sqrt[m]{b} + \sqrt[m]{a} \right)^m e^{c(n)+1/20} \right)}{\log \left(\left(\sqrt[m]{b} - \sqrt[m]{a} \right)^m e^{c(n)} \right)} \right\}.$$

Proof (sketch). To simplify the exposition of the proof, we assume that $b = a + 1$ and $a \geq 2$, especially because we aim to use this theorem for values of a and b satisfying these conditions.

We will apply Theorem 6.1 with $\theta = \left(\frac{a+1}{a} \right)^{1/n}$, $k = m - 1$ and

$$P_{i-1, r} = \sum_{j=1}^m a^r \Delta_{m, n, r} A_{ij} \left(-\frac{1}{a}, r \right) x^{j-1}.$$

From Lemma 6.2 and Proposition 6.4 it follows that

$$\begin{aligned} |P_{i, r}(\theta)| &\leq \frac{e^{c(n)r+d(n)}}{(m-1)! a^m} \left((a+1)^{1/m} - a^{1/m} \right)^{mr} \\ &\leq \frac{e^{d(n)}}{(m-1)! \cdot 2^m} \cdot \left(e^{c(n)} \left((a+1)^{1/m} - a^{1/m} \right)^m \right)^r, \end{aligned}$$

so we can choose

$$D = e^{-c(n)} \left((a+1)^{1/m} - a^{1/m} \right)^{-m}$$

and an upper bound of $\frac{e^{d(n)}}{(m-1)! \cdot 2^m}$ as d . Running through all the possible values, we see that we can take $d = e^{17.75}$.

Using Lemma 6.3 and Proposition 6.4, we obtain

$$|a_{ij}| \leq 2(r+1) \Phi_{m, n, r}^{m-1} e^{c(n)r+d(n)} \left((a+1)^{1/m} + a^{1/m} \right)^{mr}.$$

Now, according to [2], the following are valid choices for c and C :

$$c = 60.38n^{m-1}e^{d(n)},$$

$$C = e^{c(n)+\frac{1}{20}} \left((a+1)^{1/m} + a^{1/m} \right)^m.$$

If $\max\{|\theta|, |p/q|\} < 3^{1/3}$ the conclusion can be obtained by taking $r_0 = 1$ and $t = \frac{n}{n-m+1}$ in Lemma 6.1, and otherwise the conclusion follows from Liouville's theorem. $\square$

Remark. Proving Theorem 4.1 for general (a, b) with $a < b$ reduces to the case $a/b \leq 3$. Our proof is essentially the proof from [2] particularized to the case $b = a + 1$. The cases with $a/b > 3$ can be tackled with Liouville's Theorem.

Remark. Verifying that c and C as chosen above are valid comes down to proving the inequality

$$30.19 \frac{e^{r/20}}{r+1} \geq \max \left\{ \frac{n}{n-1} r^{1/n}, \frac{n}{(m-1)(n-m+1)} r^{\frac{m-1}{n}} \right\}.$$

Since we have

$$\frac{n}{n-1} r^{\frac{1}{n}} \leq \frac{17}{16} r^{\frac{1}{17}}$$

and

$$\frac{n}{(m-1)(n-m+1)} r^{\frac{m-1}{n}} \leq r^{\frac{1}{3}},$$

it would be enough to prove the inequalities

$$30.19 \frac{e^{r/20}}{r+1} \geq \frac{17}{16} r^{\frac{1}{17}},$$

$$30.19 \frac{e^{r/20}}{r+1} \geq r^{\frac{1}{3}}.$$

This is not difficult after using the inequality $e^x > 1 + x + \frac{x^2}{2}$ (for $x > 0$).

7 The equation $Ax^3 + By^3 = 1$

In 1922, Delone [10] proved that if d is an integer, then the equation

$$x^3 + dy^3 = 1$$

has at most one solution in nonzero integers. Three years later, Nagell [22] proved the same result independently and also generalized it by proving that for any integers A and B , the equation

$$Ax^3 + By^3 = 1$$

has at most one solution in nonzero integers. He also obtained a related result about the equation $Ax^3 + By^3 = 3$ (see Chapter 8).

Paralleling Nagell's approach, Ljunggren [17] found that each of the equations

$$|ax^4 - by^4| = c,$$

with $a, b > 0$ and $c \in \{1, 2, 4, 8\}$ has at most one solution in positive integers x and y .

In this chapter we will delve into Nagell's proof in the case $|A|, |B| \neq 1$. The other case does not differ substantially, so we do not insist on it.

We first provide some algebraic number theory background focused on the case of real cubic fields. We then prove some important lemmas about the coefficients appearing in powers of algebraic numbers of certain forms. Finally, we combine all these ingredients to obtain the main result.

7.1 Overview of real cubic fields

The results that we present in this section are standard, so we omit their proofs. The interested reader can find the proofs in most books on algebraic number theory, such as [23] or [28].

A complex number that is the root of a nonzero polynomial with rational coefficients is called an *algebraic number*. A complex number that is the root of a monic polynomial with integer coefficients is called an *algebraic integer*.

A *number field* $K \subset \mathbb{C}$ is a finite extension of $\mathbb{Q}$. The set of algebraic integers in K forms a subring of K , which is denoted by $\mathcal{O}_K$ and is called the *ring of (algebraic) integers* of K .

A set of algebraic integers $\{\alpha_1, \dots, \alpha_k\}$ is called an *integral basis* of K (or $\mathcal{O}_K$) if every element of $\mathcal{O}_K$ can be represented uniquely in the form

$$n_1\alpha_1 + \dots + n_k\alpha_k,$$

where $n_1, \dots, n_k$ are integers.

From now on, we assume that K is a real cubic field, that is $K = \mathbb{Q}(\sqrt[3]{D})$, with $D > 1$ an integer not divisible by the cube of any prime number. We say that K is *of the first kind* if $D \not\equiv 1 \pmod{9}$ and K is *of the second kind* if $D \equiv \pm 1 \pmod{9}$.

Let us put $D = AB^2$, with A, B coprime positive integers, not divisible by the cube of any prime. Then an integral basis of K is given by

$$\left\{1, \sqrt[3]{AB^2}, \sqrt[3]{A^2B}\right\},$$

if K is of the first kind, and by

$$\left\{\frac{1}{3} \left(1 + A\sqrt[3]{AB^2} + B\sqrt[3]{A^2B}\right), \sqrt[3]{AB^2}, \sqrt[3]{A^2B}\right\},$$

if K is of the second kind. A proof of this result can be found in [21, Example 4.3.6]. Using this, one can show the following proposition.

Proposition. *Let x, y, z be integers such that*

$$\frac{1}{3} \left(x + y\sqrt[3]{AB^2} + z\sqrt[3]{A^2B}\right)$$

is an algebraic integer. If K is of the first kind, then x, y, z are divisible by 3, and if K is of the second kind and one of the numbers x, y, z is divisible by 3, then so are the others.

There are three embeddings of K in $\mathbb{C}$, given by sending $\sqrt[3]{D}$ respectively in $\sqrt[3]{D}$, $\rho\sqrt[3]{D}$ and $\rho^2\sqrt[3]{D}$, where ρ is a primitive third root of unity. Written explicitly, these are $\sigma_1, \sigma_2, \sigma_3 : K \rightarrow \mathbb{C}$,

$$\begin{aligned}\sigma_1 \left(x + y\sqrt[3]{D} + z\sqrt[3]{D^2}\right) &= x + y\sqrt[3]{D} + z\sqrt[3]{D^2}, \\ \sigma_2 \left(x + y\sqrt[3]{D} + z\sqrt[3]{D^2}\right) &= x + y\rho\sqrt[3]{D} + z\rho^2\sqrt[3]{D^2}, \\ \sigma_3 \left(x + y\sqrt[3]{D} + z\sqrt[3]{D^2}\right) &= x + y\rho^2\sqrt[3]{D} + z\rho\sqrt[3]{D^2},\end{aligned}$$

for all $x, y, z \in \mathbb{Q}$.

For an element $\alpha \in K$ we call the elements $\sigma_1(\alpha)$, $\sigma_2(\alpha)$ and $\sigma_3(\alpha)$ the *conjugates* of α and we define the *norm* of α , denoted $N(\alpha)$, to be their product. The map N thus defined is multiplicative. An explicit description of it is given by

$$N \left(x + y\sqrt[3]{D} + z\sqrt[3]{D^2}\right) = x^3 + Dy^3 + D^2z^3 - 3Dxyz.$$

An element $\alpha \in \mathcal{O}_K$ is a unit if and only if $N(\alpha) = \pm 1$. Notice that $\sigma_2(\alpha)\sigma_3(\alpha) = \sigma_2(\alpha)\overline{\sigma_2(\alpha)} = |\sigma_2(\alpha)|^2 \geq 0$, so in case α is positive, then it is a unit if and only if $N(\alpha) = 1$.

From a classical theorem of Dirichlet we know that there is a unit $\xi \in \mathcal{O}_K$ such that any unit of $\mathcal{O}_K$ can be uniquely represented in the form $\pm\xi^k$, with k an integer. We call any such unit a *fundamental unit* of K . Notice that if ξ is a fundamental unit then so are $\pm\xi$ and $\pm 1/\xi$, and exactly one of these lies in the interval $(0, 1)$.

7.2 Powers of algebraic numbers of certain forms

We list some important lemmas from [22]. Due to the similarity of their proofs, we limit ourselves to proving only Lemma 7.4. We mention that the Lemmas 7.1 and 7.2 are not actually needed in the case we analyze, but they are used in the case $B = 1$, so we decided to include them for completeness.

Lemma 7.1. *Let $D > 1$ be an integer which is not divisible by the cube of any prime number. Let x and y be two integers such that x and yD are coprime. In n is a positive integer and X, Y, Z are integers such that*

$$\left(x + y\sqrt[3]{D}\right)^n = X + Y\sqrt[3]{D} + Z\left(\sqrt[3]{D}\right)^2,$$

then $XYZ \neq 0$, except in the following cases:

$$\begin{aligned}\left(\sqrt[3]{10} - 1\right)^5 &= 99 - 45\sqrt[3]{10}, \\ \left(\sqrt[3]{4} - 1\right)^4 &= -15 + 12\sqrt[3]{2}.\end{aligned}$$

Lemma 7.2. *Let $D_1, D_2 > 1$ be two integers that are not divisible by the cube of any prime number. Let x and y be two integers and let n be a positive integer. If xD_1 and yD_2 are coprime and X, Y, Z are integers such that*

$$\left(x\sqrt[3]{D_1} + y\sqrt[3]{D_2}\right)^{3n} = X + Y\sqrt[3]{D_1D_2^2} + Z\sqrt[3]{D_1^2D_2},$$

then $XYZ \neq 0$, except in the following case:

$$\left(\sqrt[3]{5} - \sqrt[3]{3}\right)^6 = -171 + 63\sqrt[3]{20}.$$

Lemma 7.3. *Let $D_1, D_2 > 1$ be two integers that are not divisible by the cube of any prime number. Let x and y be two integers and let n be a positive integer. If xD_1 and yD_2 are coprime and X, Y, Z are integers such that*

$$\left(x\sqrt[3]{D_1} + y\sqrt[3]{D_2}\right)^{3n-1} = X\left(\sqrt[3]{D_1}\right)^2 + Y\left(\sqrt[3]{D_2}\right)^2 + Z\sqrt[3]{D_1D_2},$$

then $XYZ \neq 0$.

Lemma 7.4. *Let $D_1, D_2 > 1$ be two integers that are not divisible by the cube of any prime number. Let x and y be two integers and let n be a positive integer. If xD_1 and yD_2 are coprime and X, Y, Z are integers such that*

$$\left(x\sqrt[3]{D_1} + y\sqrt[3]{D_2}\right)^{3n+1} = X\sqrt[3]{D_1} + Y\sqrt[3]{D_2} + Z\left(\sqrt[3]{D_1D_2}\right)^2,$$

then $XYZ \neq 0$.

Proof of Lemma 7.4. If $X = 0$, we get

$$\sum_{k=0}^n \binom{3n+1}{3k} x^{3(n-k)+1} y^{3k} D_2^k = 0,$$

which is impossible, because the left hand side is not divisible by D_2 (the first term of the sum is not divisible by D_2 , while the rest of the terms are). By a symmetric argument, we find $Y \neq 0$.

The case $Z = 0$ leads to the equation

$$\sum_{k=0}^{n-1} \binom{3n+1}{3k+2} x^{3n-3k-1} y^{3k+2} D_1^{n-k-1} D_2^k = 0. \quad (7.1)$$

Dividing by $\frac{3}{2}n(3n+1)x^2y^2$ and rearranging, we get

$$-x^{3n-3}D_1^{n-1} = 2 \sum_{k=1}^{n-1} \binom{3n-1}{3k} \frac{x^{3n-3k-3}y^{3k}D_1^{n-k-1}D_2^k}{(3k+1)(3k+2)}. \quad (7.2)$$

Let's assume that $|y| \neq 1$ and consider a prime divisor q of y . Notice that for each $k > 1$, the numbers $3k+1$ and $3k+2$ are coprime and smaller than q^{3k} , so

$$v_q((3k+1)(3k+2)) < 3k.$$

Therefore, the q -adic valuation of the summand in (7.2) is positive, which implies that the right hand side is divisible by q . On the other hand, since xD_1 and yD_2 are coprime, q does not divide the left hand side, which is a contradiction, so $|y| = 1$.

Similarly, one can show that if r^α divides D_2 and $r^\alpha \notin \{2, 5\}$, with r prime and $\alpha \in \{1, 2\}$, then the right hand side of (7.2) is divisible by r , which is again a contradiction. It follows that $D_2 \in \{2, 5, 10\}$.

Due to the symmetry of (7.1) with respect to the pairs of variables (x, y) and (D_1, D_2) , we arrive at the same conclusions about y and D_1 , that is $|y| = 1$ and $D_1 \in \{2, 5, 10\}$.

Considering the symmetry and the fact that D_1 and D_2 are coprime, it remains to prove the lemma for $D_1 = 5$ and $D_2 = 2$. From (7.1) it clearly follows that x and y must have opposite signs, so we may assume without loss of generality $x = 1$ and $y = -1$.

Proving the lemma now comes down to proving that the coefficient of $\sqrt[3]{10}$ in the expansion of

$$\left(\sqrt[3]{5} - \sqrt[3]{2}\right)^{3n+1}$$

is nonzero.

We now consider two cases, based on the parity of n . Let us assume that n is odd, say $n = 2s + 1$, with s a nonnegative integer. Then

$$\begin{aligned} \left(\sqrt[3]{5} - \sqrt[3]{2}\right)^{3s+2} &= \left(\sqrt[3]{5} - \sqrt[3]{2}\right)^{3s} \left(\sqrt[3]{5} - \sqrt[3]{2}\right)^2 \\ &= \left(3 + 3\sqrt[3]{20} - 3\sqrt[3]{50}\right)^s \left(\sqrt[3]{25} - 2\sqrt[3]{10} + \sqrt[3]{4}\right) \\ &= 3^s \left(x\sqrt[3]{25} + y\sqrt[3]{10} + z\sqrt[3]{4}\right), \end{aligned}$$

where x, y, z are integers. Notice that by multiplying this last relation by $(\sqrt[3]{4})^{3s+2}$ we get

$$\left(\sqrt[3]{20} - 2\right)^{3s+2} = 12^s \left(x\sqrt[3]{400} + 2y\sqrt[3]{20} + 4z\right),$$

which is an equality in $\mathbb{Q}(\sqrt[3]{20})$, so applying the corresponding norm yields

$$12^{3s+2} = 12^{3s} (400x^3 + 160y^3 + 64z^3 - 480xyz),$$

and so

$$25x^3 + 10y^3 + 4z^3 - 30xyz = 9. \quad (7.3)$$

From the assumption $Z = 0$ we have

$$\left(\sqrt[3]{5} - \sqrt[3]{2}\right)^{3n+1} = X\sqrt[3]{5} + Y\sqrt[3]{2}, \quad (7.4)$$

so

$$9^s \left(x\sqrt[3]{25} + y\sqrt[3]{10} + z\sqrt[3]{4}\right)^2 = X\sqrt[3]{5} + Y\sqrt[3]{2}.$$

By equating the coefficient of $\sqrt[3]{10}$ on the left hand side with 0, we get

$$y^2 = -2xz. \quad (7.5)$$

If $d = \gcd(x, z)$, the last equality implies that d^2 divides y^2 , so d divides y . That is, d is a common divisor for x , y and z . On the other hand, from (7.3) we find that d^3 divides 9, which is only possible if $d = 1$. Relation (7.3) also implies that x is odd, so using this and the coprimality of x and z in (7.5) allows us to conclude that x , y and z have the form

$$(x, y, z) = (\pm u^2, 2uv, \mp 2v^2),$$

where u and v are coprime integers.

Now notice that (7.3) implies $z^3 \equiv 1 \pmod{5}$, and so $\mp 3v^6 \equiv 1 \pmod{5}$, which implies $v^6 \equiv \mp 2 \pmod{5}$. However, 2 and -2 are not quadratic residues modulo 5, so the last congruence is impossible.

We now assume that n is even, say $n = 2s$, with s a positive integer. Relation (7.4) becomes

$$\left(-171 + 63\sqrt[3]{20}\right)^s \left(\sqrt[3]{5} - \sqrt[3]{2}\right) = X\sqrt[3]{5} + Y\sqrt[3]{2}.$$

The coefficient of $\sqrt[3]{100}$ on the left hand side is

$$\begin{aligned} & \sum_{k=0}^{\lfloor \frac{s-1}{3} \rfloor} \binom{s}{3k+1} (-171)^{s-3k-1} \cdot 63^{3k+1} \cdot 20^k - \\ & - 2 \sum_{k=0}^{\lfloor \frac{s-2}{3} \rfloor} \binom{s}{3k+2} (-171)^{s-3k-2} \cdot 63^{3k+2} \cdot 20^k. \end{aligned}$$

Equating it with 0 yields

$$\begin{aligned} (-19)^{s-1} + \sum_{k=0}^{\lfloor \frac{s-1}{3} \rfloor} \binom{s-1}{3k} \frac{(-171)^{s-3k-1} \cdot 7^{3k} \cdot 20^k}{3k+1} = \\ = 2 \sum_{k=0}^{\lfloor \frac{s-2}{3} \rfloor} \binom{s-1}{3k+1} \frac{(-171)^{s-3k-2} \cdot 7^{3k+1} \cdot 20^k}{3k+2}. \end{aligned}$$

However, it is clear that the right hand side is divisible by 7, while the left hand side is not. This contradiction finishes the proof. $\square$

Remark. *The proof of Lemma 7.3 is analogous to the beginning of the proof of Lemma 7.4. In this case one gets $D_1, D_2 \in \{2, 4\}$ which finishes the proof with a contradiction (as D_1 and D_2 were supposed to be coprime).*

7.3 The main result

It is clear that we do not lose the generality by assuming $A > B > 1$ and that A and B are not divisible by the cube of any prime. The equation forces A and B to be coprime (otherwise there are no solutions). Let $A = ac^2$ and $B = bd^2$, where the numbers a, b, c, d are positive integers. Clearly a, b, c, d are pairwise coprime and not divisible by the cube of any prime. We consider $K = \mathbb{Q}(\sqrt[3]{AB^2})$.

Solving the equation $Ax^3 + By^3 = 1$ turns out to be a chase of units in $\mathcal{O}_K$. Indeed, the equation can be written as

$$\left(x\sqrt[3]{A} + y\sqrt[3]{B}\right) \left(x\sqrt[3]{A} + y\rho\sqrt[3]{B}\right) \left(x\sqrt[3]{A} + y\rho^2\sqrt[3]{B}\right) = 1.$$

Cubing this relation shows that the product of the conjugates of

$$\left(x\sqrt[3]{A} + y\sqrt[3]{B}\right)^3 = 1 + 3x^2y\sqrt[3]{A^2B} + 3xy^2\sqrt[3]{AB^2}$$

in K is 1, so this element has norm 1 in K , and so it is a positive unit in $\mathcal{O}_K$.

Now, from the equality

$$x\sqrt[3]{A} + y\sqrt[3]{B} = \frac{1}{x^2\sqrt[3]{A^2} - xy\sqrt[3]{AB} + y^2\sqrt[3]{B^2}}$$

and the observation that x and y have different signs, it follows

$$x\sqrt[3]{A} + y\sqrt[3]{B} < \frac{1}{3}.$$

Therefore, if ξ is a fundamental unit of K , it follows that

$$\left(x\sqrt[3]{A} + y\sqrt[3]{B}\right)^3$$

is a positive power of ξ . The following result gives a more explicit description.

Theorem 7.5. *If ξ is the fundamental unit of K lying in the interval $(0, 1)$, and m is an integer such that*

$$\left(x\sqrt[3]{A} + y\sqrt[3]{B}\right)^3 = \xi^m,$$

then m is a power of two.

Proof. Proving the theorem comes down to showing that if μ is a unit in $\mathcal{O}_K$ and p is an odd prime, then the equality

$$\left(x\sqrt[3]{A} + y\sqrt[3]{B}\right)^3 = \mu^p \tag{7.6}$$

is impossible.

If $p = 3$, then (7.6) implies

$$x\sqrt[3]{A} + y\sqrt[3]{B} = \mu,$$

which is impossible, because μ is an algebraic number of degree 3, while the left hand side is an algebraic number of degree 9.

Now let's assume that $p = 3m + 1$, with m a positive integer. Relation (7.6) implies

$$\sqrt[3]{\mu} = \left(x\sqrt[3]{A} + y\sqrt[3]{B}\right) \mu^{-m}.$$

The element μ^{-m} , as an algebraic integer in $\mathcal{O}_K$, is of the form

$$\frac{1}{3} \left(X + Y\sqrt[3]{ac^2b^2d} + Z\sqrt[3]{a^2cbd^2} \right),$$

where X, Y, Z are integers, so it follows that

$$\varepsilon := \sqrt[3]{\mu} = \frac{1}{3} \left(x_1\sqrt[3]{ac^2} + y_1\sqrt[3]{bd^2} + z_1\sqrt[3]{a^2cb^2d} \right), \tag{7.7}$$

with x_1, y_1, z_1 integers.

Since μ is a unit in $\mathcal{O}_K$, using a norm argument yields

$$\frac{1}{3^9} \left(x_1^3ac^2 + y_1^3bd^2 + z_1^3a^2cb^2d - 3x_1y_1z_1abcd \right)^3 = 1,$$

so

$$x_1^3ac^2 + y_1^3bd^2 + z_1^3a^2cb^2d - 3x_1y_1z_1abcd = 27. \tag{7.8}$$

From (7.6) and (7.7) we get

$$\left(\frac{1}{3} \left(x_1\sqrt[3]{ac^2} + y_1\sqrt[3]{bd^2} + z_1\sqrt[3]{a^2cb^2d} \right) \right)^p = x\sqrt[3]{ac^2} + y\sqrt[3]{bd^2}. \tag{7.9}$$

Multiplying this relation by $\left(\sqrt[3]{a^2c}\right)^p$, we obtain an equality in $\mathbb{Q}\left(\sqrt[3]{a^2bcd^2}\right)$, to which we apply the complex embeddings of this field and divide back by $\left(\sqrt[3]{a^2c}\right)^p$, thus getting

$$\left(\frac{1}{3}\left(x_1\sqrt[3]{ac^2} + y_1\rho\sqrt[3]{bd^2} + z_1\rho^2\sqrt[3]{a^2cb^2d}\right)\right)^p = x\sqrt[3]{ac^2} + y\rho\sqrt[3]{bd^2} \quad (7.10)$$

and

$$\left(\frac{1}{3}\left(x_1\sqrt[3]{ac^2} + y_1\rho^2\sqrt[3]{bd^2} + z_1\rho\sqrt[3]{a^2cb^2d}\right)\right)^p = x\sqrt[3]{ac^2} + y\rho^2\sqrt[3]{bd^2}, \quad (7.11)$$

where ρ is a primitive cube root of unity.

Multiplying (7.10) by ρ and (7.11) by ρ^2 and then adding the resulted relations, keeping in mind that $\rho = \rho^p$ and $\rho^2 + \rho + 1 = 0$, yields

$$\begin{aligned} &\left(\frac{x_1\rho\sqrt[3]{ac^2} + y_1\rho^2\sqrt[3]{bd^2} + z_1\sqrt[3]{a^2cb^2d}}{3}\right)^p + \left(\frac{x_1\rho^2\sqrt[3]{ac^2} + y_1\rho\sqrt[3]{bd^2} + z_1\sqrt[3]{a^2cb^2d}}{3}\right)^p \\ &= -x\sqrt[3]{ac^2} - y\sqrt[3]{bd^2}. \end{aligned}$$

The right hand side of the above relation is equal to $-\varepsilon^p$, according to (7.9), and the left hand side is divisible by

$$\begin{aligned} \varepsilon' &:= \frac{x_1\rho\sqrt[3]{ac^2} + y_1\rho^2\sqrt[3]{bd^2} + z_1\sqrt[3]{a^2cb^2d}}{3} + \frac{x_1\rho^2\sqrt[3]{ac^2} + y_1\rho\sqrt[3]{bd^2} + z_1\sqrt[3]{a^2cb^2d}}{3} \\ &= -\frac{x_1\sqrt[3]{ac^2} + y_1\sqrt[3]{bd^2} - 2z_1\sqrt[3]{a^2cb^2d}}{3}. \end{aligned}$$

Notice that ε' is the sum of two algebraic integers, so it is algebraic integer. Since it divides ε^p , it follows that ε' is a unit in the ring of integers of $\mathbb{Q}(\sqrt[3]{ac^2}, \sqrt[3]{bd^2})$, so $(\varepsilon')^3$ is a unit in $\mathcal{O}_K$. Using a norm argument we find that

$$x_1^3ac^2 + y_1^3bd^2 - 8z_1^3a^2cb^2d + 6x_1y_1z_1abcd = \pm 27.$$

Combining this with (7.8) yields

$$abcdz_1(cdz_1^2 - x_1y_1) = 6 \text{ or } 0.$$

The first case implies that a, b, c, d are not all divisible by 3, so the field K is of the second kind, whence $ac^2bd^2 \equiv \pm 1 \pmod{9}$. But $abcd$ can only be 2, 3 or 6, and it is easy to see that the congruence is false in each case.

The second case implies $z_1 = 0$ or $cdz_1^2 = x_1y_1$.

If $z_1 = 0$, then (7.9) becomes

$$\left(\frac{x_1}{3}\sqrt[3]{ac^2} + \frac{y_1}{3}\sqrt[3]{bd^2}\right)^p = x\sqrt[3]{ac^2} + y\sqrt[3]{bd^2}, \quad (7.12)$$

so

$$\frac{x_1}{3}\sqrt[3]{ac^2} + \frac{y_1}{3}\sqrt[3]{bd^2}$$

is an algebraic integer.

If K is of the first kind, then $x_1/3$ and $x_2/3$ are integers. If K is of the second kind, by noticing that the number

$$\sqrt[3]{a^2c} \left(\frac{x_1}{3} \sqrt[3]{ac^2} + \frac{y_1}{3} \sqrt[3]{bd^2} \right) = \frac{x_1ac}{3} + \frac{y_1}{3} \sqrt[3]{a^2cbd^2}$$

is an algebraic integer (as the product of two algebraic integers), we can conclude that $x_1ac/3$ and $y_1/3$ are integers. The fact that K is of the second kind also implies that ac is coprime to 3, and so $x_1/3$ is an integer.

Both cases led to the conclusion that $x_1/3$ and $y_1/3$ are integers. Moreover, using $z_1 = 0$ in (7.8), we conclude that they are coprime. Applying Lemma 7.4, we conclude that equality (7.12) is actually impossible.

In the case $cdz_1^2 = x_1y_1$ one easily sees that $1/\varepsilon = x_2\sqrt[3]{a^2c} + y_2\sqrt[3]{b^2d}$, for some rational numbers x_2 and y_2 . Similar considerations as before lead to the conclusion that x_2 and y_2 are in fact integers and they satisfy $x_2^3a^2c + y_2^3b^2d = 1$, so they have opposite signs. This leads to

$$\varepsilon = x_2^2\sqrt[3]{a^4c^2} - x_2y_2\sqrt[3]{a^2cb^2d} + y_2^2\sqrt[3]{b^4d^2} > 3,$$

contradicting the fact that $\varepsilon < 1$.

The case $p \equiv 2 \pmod{3}$ is analogous. □

Now that the theorem is proved, we can finally establish the main result of this chapter, namely that the equation $Ax^3 + By^3 = 1$ has at most one solution in integers. Assuming that there are at least two solutions, say (x_1, y_1) and (x_2, y_2) , they produce the units $(x_i\sqrt[3]{A} + y_i\sqrt[3]{B})^3$ in $\mathcal{O}_K$, and using Theorem 7.5, we have

$$(x_i\sqrt[3]{A} + y_i\sqrt[3]{B})^3 = \xi^{2^{m_i}},$$

for $i \in \{1, 2\}$, where m_1 and m_2 are distinct nonnegative integers. Assuming without loss of generality $m_1 < m_2$, we get

$$(x_1\sqrt[3]{A} + y_1\sqrt[3]{B})^{3 \cdot 2^{m_2-m_1}} = (x_1\sqrt[3]{A} + y_1\sqrt[3]{B})^3,$$

so

$$(x_1\sqrt[3]{A} + y_1\sqrt[3]{B})^{2^{m_2-m_1}} = x_1\sqrt[3]{A} + y_1\sqrt[3]{B}.$$

However, the last relation is impossible according to Lemmas 7.3 and 7.4.

8 Discussion

As we saw, the study of the equation $|ax^n - by^n| = 1$ required good approximations of numbers of the form $\sqrt[n]{1 + \frac{1}{a}}$. To obtain them, we used a Liouville-type estimate as given by Lemma 5.1. Finding such estimates, either for a general irrational number or for a specific class of irrational numbers, is a general theme of interest in the context of Diophantine approximation. The results presented in this thesis can be slightly generalized to numbers of the form $\sqrt[n]{\left(1 + \frac{1}{a}\right)^k}$, as done in [6].

Regarding the special case $n = 3$, we would like to mention that Nagell also considered the equations $Ax^3 + By^3 = 3$ in [22]. He found that they admit at most one solution in nonzero integers, except for the equation $2x^3 + y^3 = 3$, which has exactly two solutions, namely $(1, 1)$ and $(4, -5)$. He also provided an algorithm for solving equations of the form $Ax^3 + By^3 = C$, with $C \in \{1, 3\}$, but unfortunately the algorithm is not practical.

In Theorem 7.5 we saw that the solutions of the equation $Ax^3 + By^3 = 1$ can be “found” in the sequence $(\xi^{2^n})_{n \geq 0}$, where ξ is the fundamental unit of $\mathbb{Q}(\sqrt[3]{AB^2})$ from the interval $(0, 1)$. As proved in [22], the result is also true for the equation $Ax^3 + By^3 = 3$ if we replace $(x\sqrt[3]{A} + y\sqrt[3]{B})^3$ by $\frac{1}{3}(x\sqrt[3]{A} + y\sqrt[3]{B})^3$, with only one exception, namely

$$\frac{1}{3}(\sqrt[3]{2} + 1)^3 = (\sqrt[3]{2} - 1)^{-1},$$

where $\sqrt[3]{2} - 1$ is the fundamental unit of $\mathbb{Q}(\sqrt[3]{2})$ from the interval $(0, 1)$.

Ljunggren [18] improved this result by showing that if (x, y) satisfies $Ax^3 + By^3 = C$, with $A, B > 1$ and $C \in \{1, 3\}$, with AB not divisible by 3 if $C = 3$, then $C^{-1}(x\sqrt[3]{A} + y\sqrt[3]{B})^3$ is either the fundamental unit of $\mathbb{Q}(\sqrt[3]{AB^2})$ from the interval $(0, 1)$, or its square, with the same exception as before. In other words, he proved that the value of m in Theorem 7.5 is either 0 or 1.

Later, Ljunggren [17] treated the case of the equations $|ax^4 - by^4| = c$, with positive integers a, b and $c \in \{1, 2, 4, 8\}$. He proved that each of them has at most one solution in positive integers x and y .

While we did not go into the details of Ljunggren’s proof, the methods he used are generalizations in quartic and quadratic fields of the methods used by Nagell that we presented in Chapter 7.

9 Popular summary

Diophantine equations are equations for which we are interested in finding integer solutions. A notable example is the Pythagorean equation

$$x^2 + y^2 = z^2,$$

which can be solved elementarily. However, by considering slight modifications of this equation, we might obtain extremely difficult equations. It is the case of the famous Fermat equation

$$x^n + y^n = z^n,$$

where $n \geq 3$ is an integer. It took over three centuries for this equation to be solved, and, needless to say, the techniques involved in solving it are very advanced.

In addition to explicitly solving a Diophantine equation, a typical goal is to find bounds for the number of solutions to equations or families of equations. One may use various techniques to achieve this, inspired from different areas of Mathematics, such as Algebra, Analysis, or Geometry.

From a more algebraic point of view, one often wishes to extend the "playground" by choosing to work over more general rings, such as

$$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$$

or

$$\mathbb{Z}[\sqrt{6}] = \{a + b\sqrt{6} : a, b \in \mathbb{Z}\}.$$

One of the advantages of considering such rings is that one can factorize expressions that could not have been factorized otherwise, for example,

$$\begin{aligned}x^2 + y^2 &= (x + yi)(x - yi), \\x^2 - 6y^2 &= (x + y\sqrt{6})(x - y\sqrt{6}).\end{aligned}$$

In this way one can find new information by using the properties of these new rings. For example, the ring $\mathbb{Z}[i]$ is in some sense similar to $\mathbb{Z}$, as there is a result similar to the fundamental theorem of arithmetic on this ring: *unique factorization*.

From a more analytic point of view, one may be interested in studying the asymptotic behavior of certain functions, most notably involving primes. An analytic approach would most often lead to finding bounds for the unknowns, thus reducing the problem to checking finitely many possible solutions. Another question of interest that can help in the study of Diophantine equations is how well irrational numbers can be approximated

by rational numbers, which is actually a topic of interest in the field of Diophantine approximation.

However, it is important to note that often times the computations required in the process of solving Diophantine equations might be impossible for a human. In this case the computer comes as a very useful tool.

In this thesis we are interested in finding the number of positive solutions of the equation

$$|ax^n - by^n| = 1,$$

with positive integers a , b and $n \geq 3$. To this end, we employ various methods, as some of the ones described above.

Appendix

We list a few results concerning the functions $\theta(x)$ and $\theta(x, k, l)$ that were used to prove Lemma 5.5. We recall the definitions of these functions:

$$\theta(x) = \sum_{p \leq x} \log p,$$

$$\theta(x; k, l) = \sum_{\substack{p \leq x \\ p \equiv l \pmod{k}}} \log p,$$

where p symbolizes a prime variable.

Theorem A ([24], Table 2). *Let $k \leq 486$ be a positive integer. For any integer l with $\gcd(k, l) = 1$ and any real number $x \in (0, 10^{10}]$ the following inequality holds:*

$$\left| \theta(x; k, l) - \frac{x - N(k, l)\sqrt{x}}{\varphi(k)} \right| \leq a_k \sqrt{x},$$

where $N(k, l)$ represents the number of solutions of the equation $t^2 = l$ in $\mathbb{Z}/k\mathbb{Z}$ and a_k is a constant given in Table 2 in [24].

In particular, for any $x \in (0, 10^{10}]$ and $l \in \{1, 2, 4\}$, the following inequality holds:

$$\left| \theta(x; 7, l) - \frac{x - 2\sqrt{x}}{6} \right| \leq 0.829249\sqrt{x},$$

whereas for $l \in \{3, 5, 6\}$ we have

$$\left| \theta(x; 7, l) - \frac{x}{6} \right| \leq 0.829249\sqrt{x}.$$

Theorem B ([24], Theorem 1). *For (k, ε, x_0) given in Table 2 of [24], and l coprime to k , we have*

$$\left| \theta(x; k, l) - \frac{x}{\varphi(k)} \right| \leq \varepsilon \cdot \frac{x}{\varphi(k)}, \text{ for } x \geq x_0.$$

In particular,

$$\left| \theta(x; 7, l) - \frac{x}{6} \right| \leq 0.003248 \cdot \frac{x}{6}, \text{ for } x \geq 10^{10}.$$

Theorem C ([26], Corollary 2*). *If c and d are constants as given in the table below, then the inequality*

$$\theta(x) > x - \frac{x}{c \log x}$$

holds for any $x \geq d$.

c	41	40	39	37	35	29	25
d	758,231	678,407	644,123	486,377	468,577	315,437	302,969
c	23	19	18	15	13	12	11
d	181,889	120,557	89,513	70,877	48,751	40,813	32,353
c	9	8	7	6	5	9/2	4
d	20,873	19,421	11,923	8,623	5,407	3,527	3,301
c	7/2	10/3	3	5/2	7/3	2	9/5
d	2,657	1,973	1,429	809	599	563	347
c	5/3	7/5	9/7	7/6	8/7	1	4/5
d	227	149	101	67	59	41	2

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